



嵌入式電腦視覺系統

賦予機器理解世界的眼睛

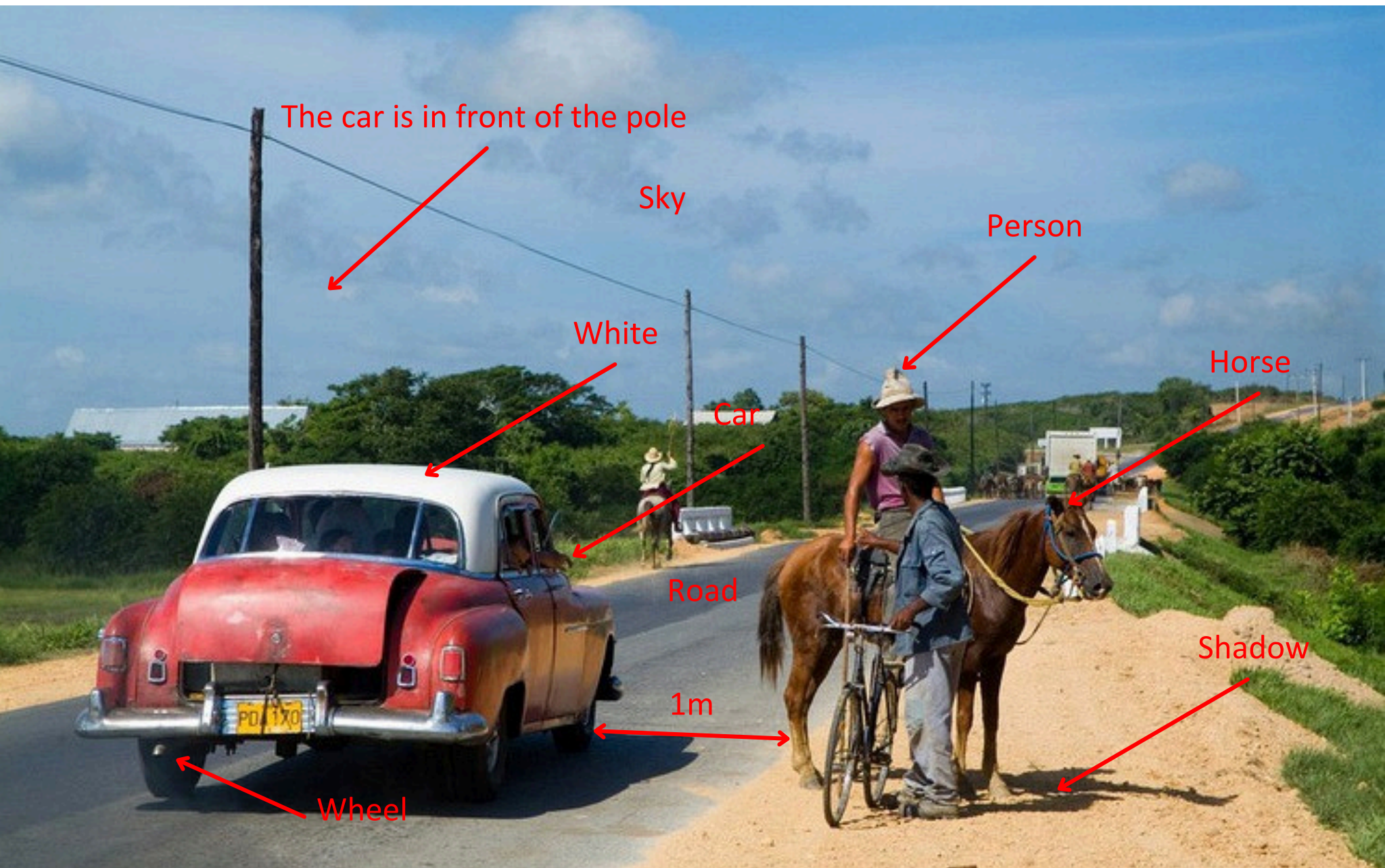
2026-03-05 謝昇憲

課程目標？

從像素中萃取意義
將視覺大腦放入手機

- 熟悉Python + OpenCV進行影像處理與電腦視覺開發
- 掌握Android 平台上的影像應用開發
- 具備機器視覺應用能力（如人臉偵測、物件追蹤、手勢識別等）
- 透過專題實作，讓學生具備開發嵌入式視覺應用的能力

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The car is in front of the pole

Sky

Person

White

Horse

Car

Road

Shadow

1m

Wheel

Computer Vision

- Low Level Vision

- Measurements
- Enhancements
- Region segmentation
- Features

- Mid Level Vision

- Reconstruction
- Depth
- Motion Estimation

- High Level Vision

- Category detection
- Activity recognition
- Deep understandings



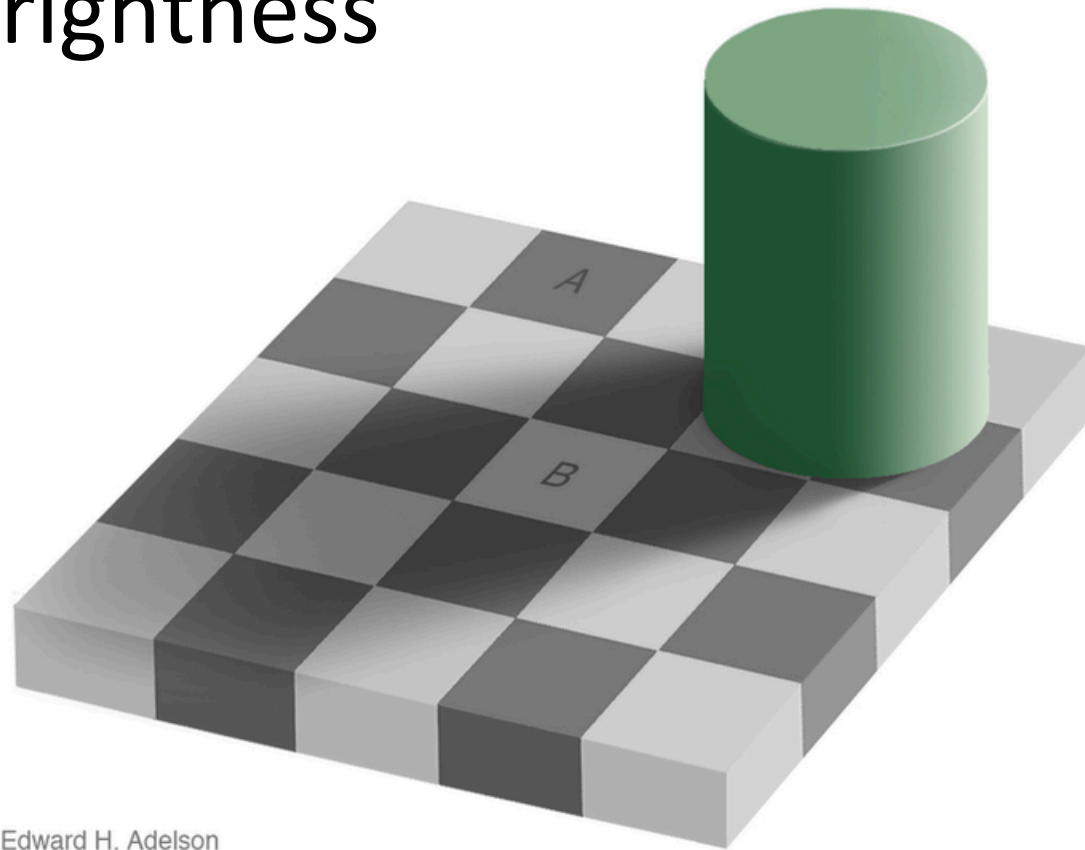
Computer Vision

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Measurement

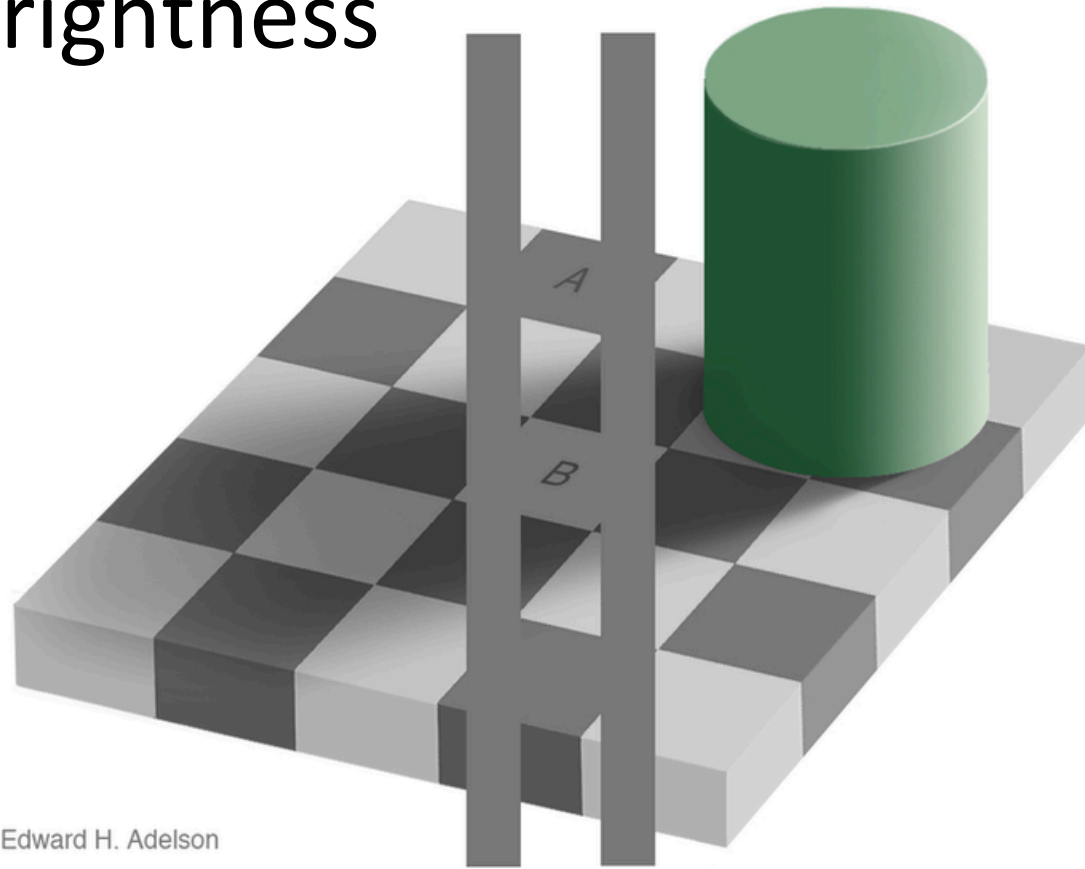
Brightness



Edward H. Adelson

Measurement

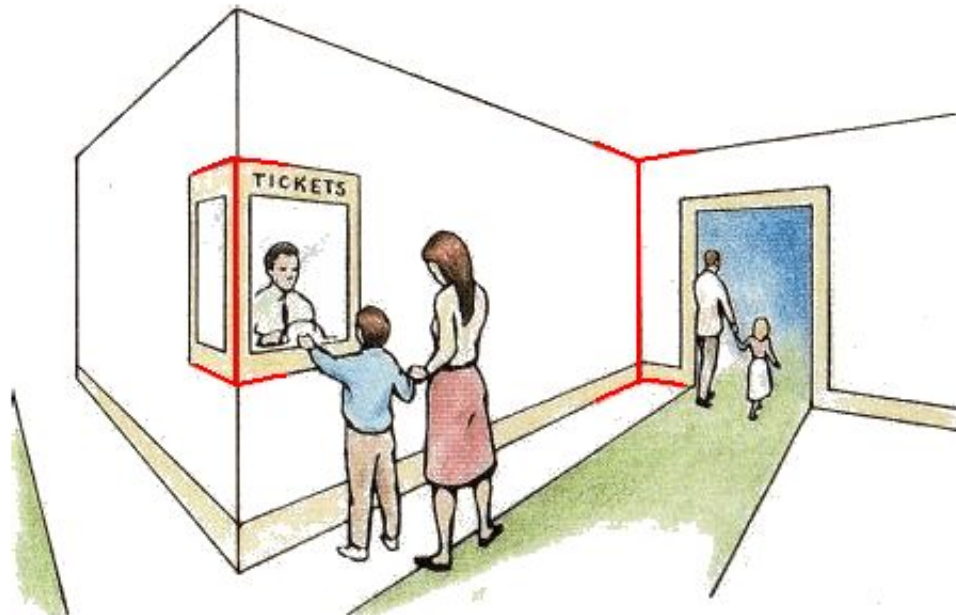
Brightness



Edward H. Adelson

Measurement

Length



Müller-Lyer Illusion

http://www.michaelbach.de/ot/sze_muelue/index.html

Image Enhancement



Image Inpainting, M. Bertalmío et al.
<http://www.iua.upf.es/~mbertalmio//restoration.html>

Image Enhancement



Image Inpainting, M. Bertalmío et al.
<http://www.iaa.upf.es/~mbertalmio//restoration.html>

Image Enhancement



Image Inpainting, M. Bertalmío et al.
<http://www.iua.upf.es/~mbertalmio//restoration.html>

Seam Carving

less
important





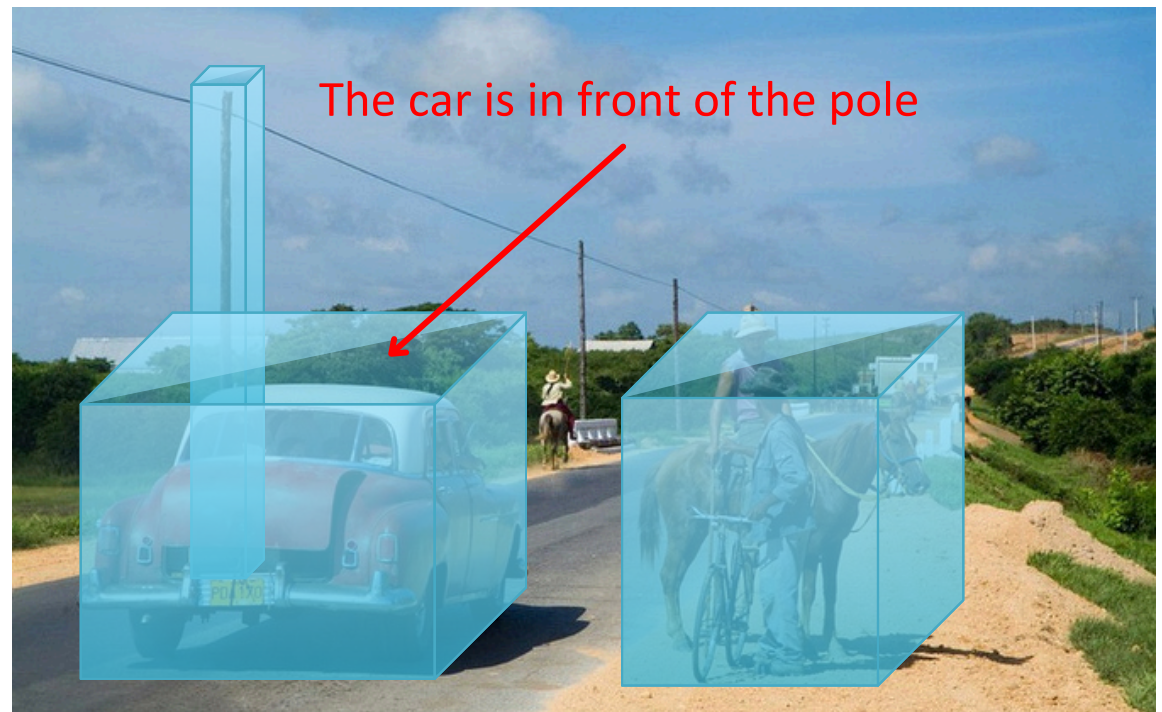
Traditional resizing uses and stretches the whole image.



**Content-aware resizing uses important areas.
Extends in horizontal direction and reduces in vertical.**

Computer Vision

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 - Deep understandings

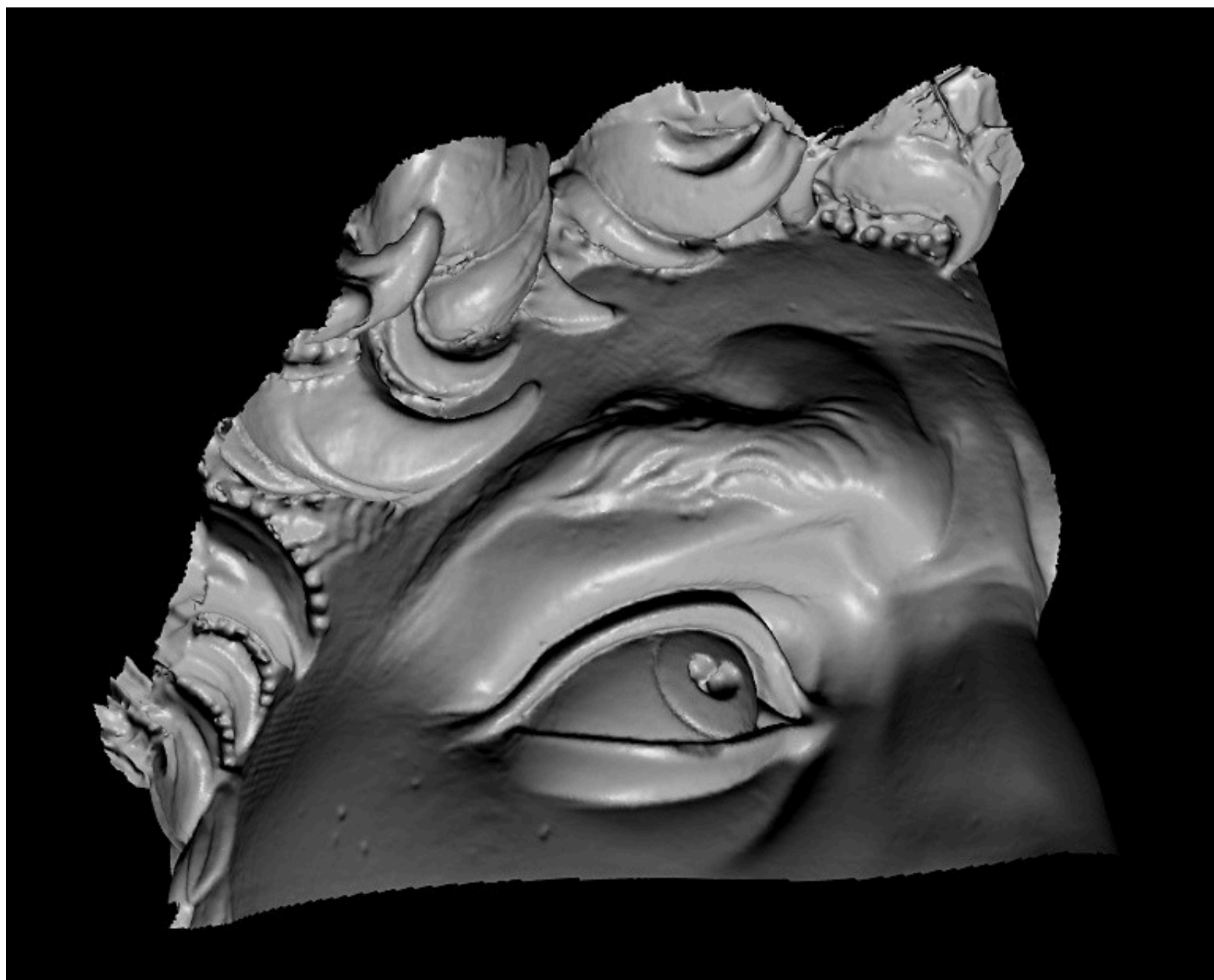


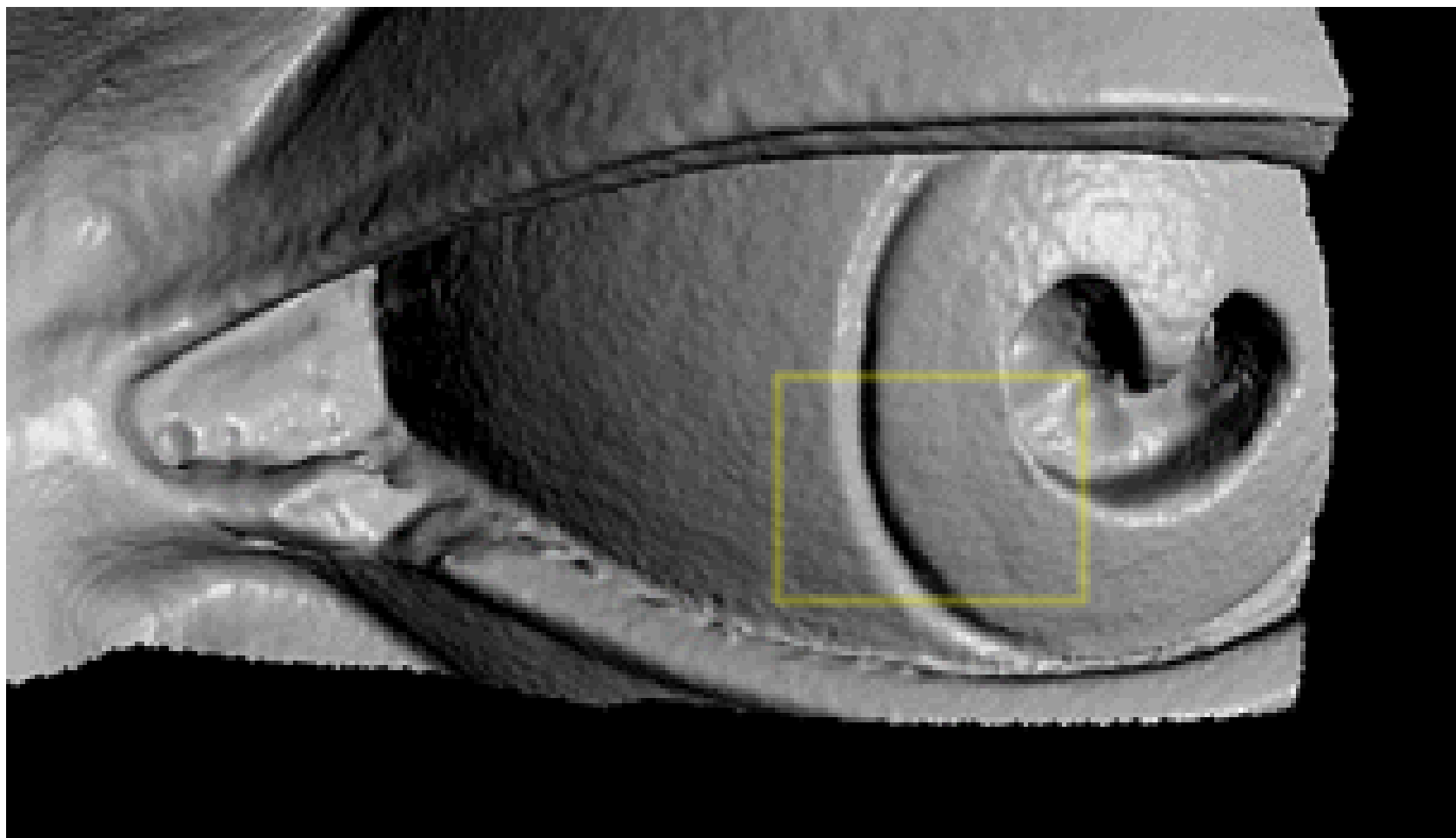
Applications: 3D Scanning

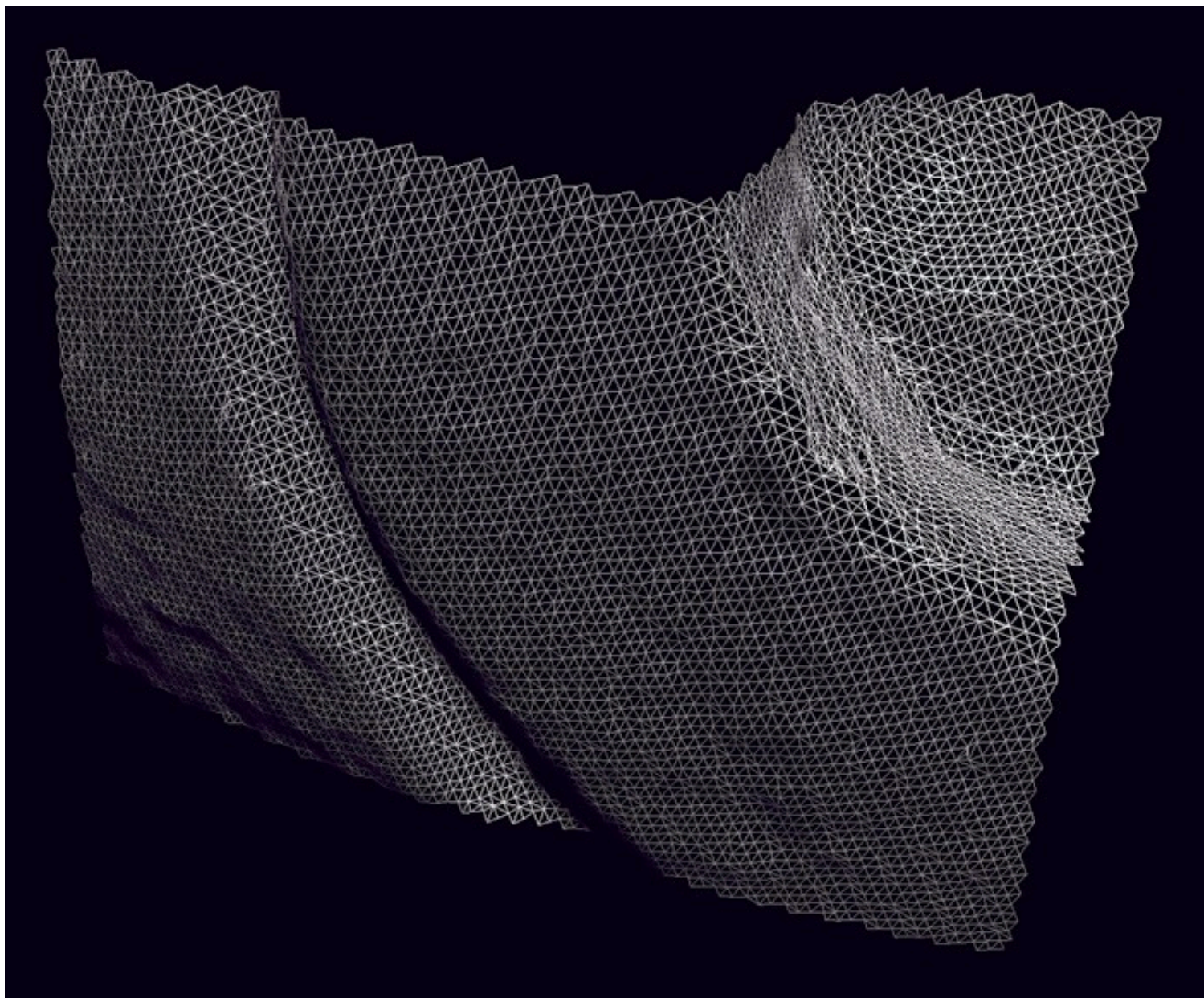


Scanning Michelangelo's *"The David"*



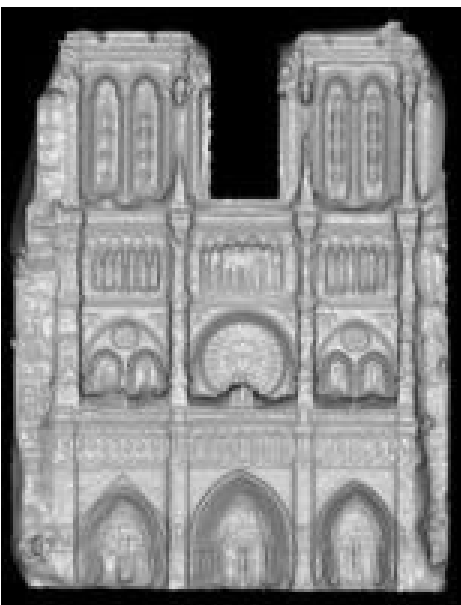
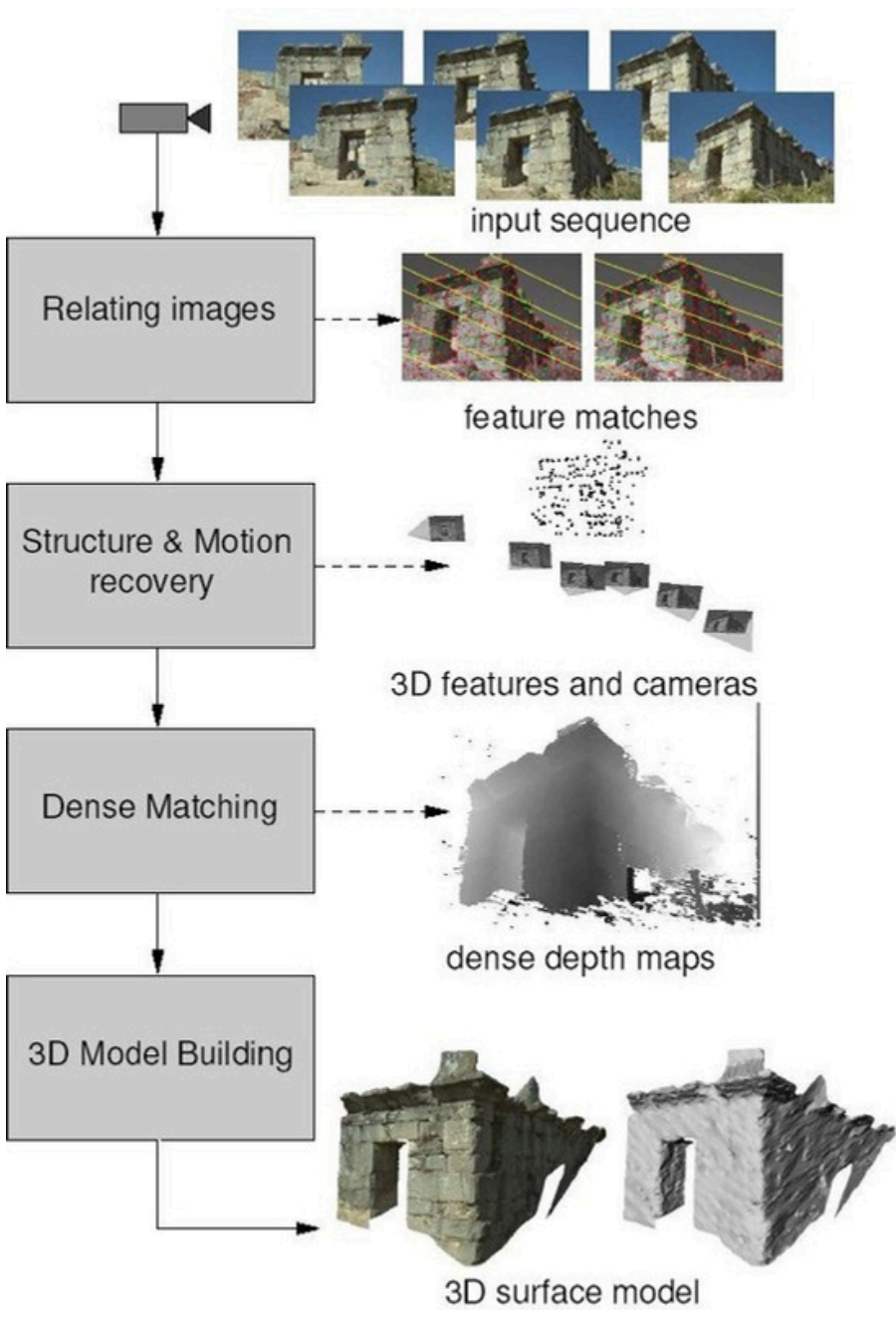


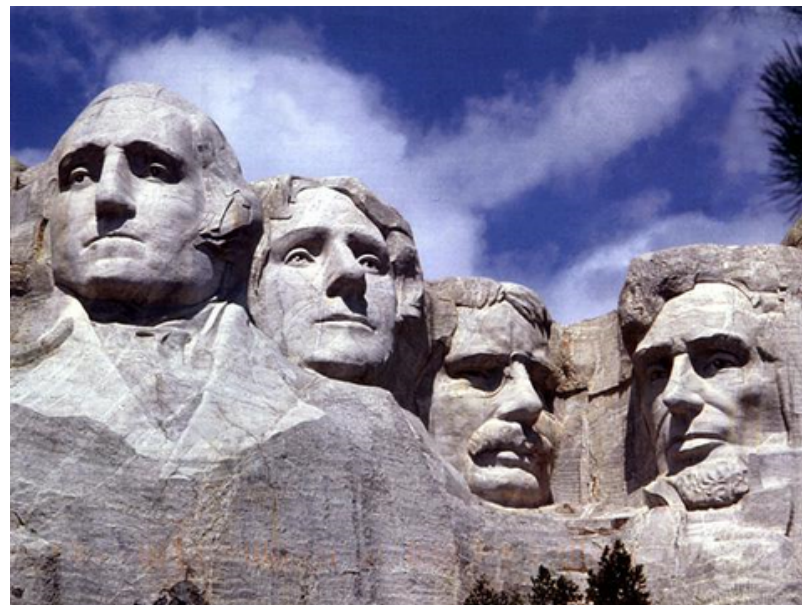




Extracting geometric information

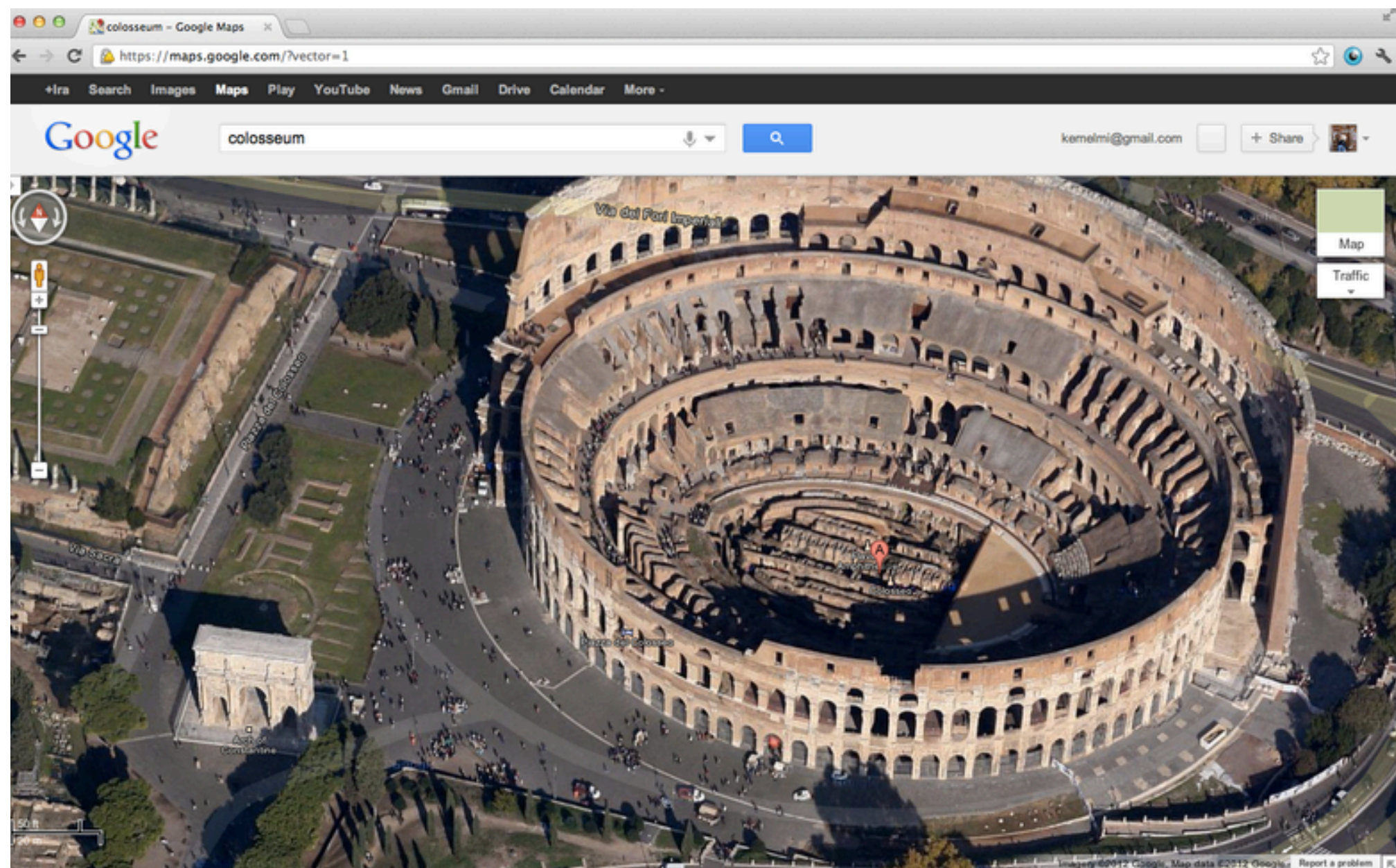
Real-time stereo





Google's 3D Maps

Structure estimation from tourist photos



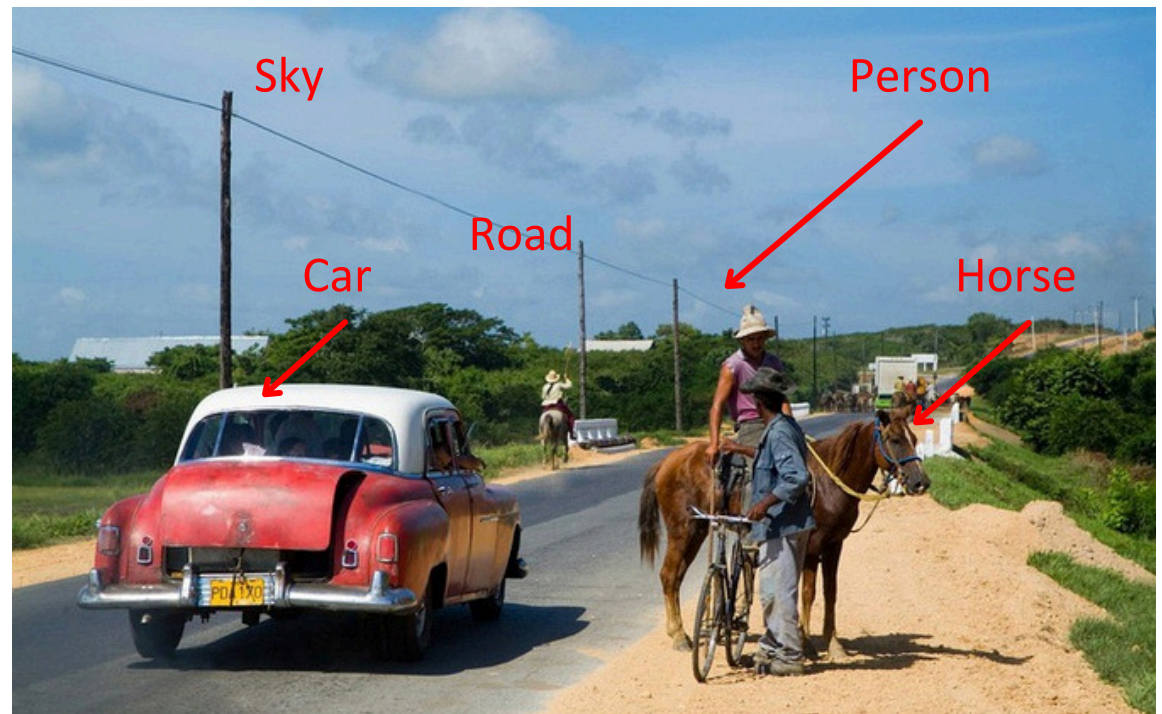
Apple's 3D maps



<https://www.youtube.com/watch?v=InIVv-LsgZE>

Computer Vision

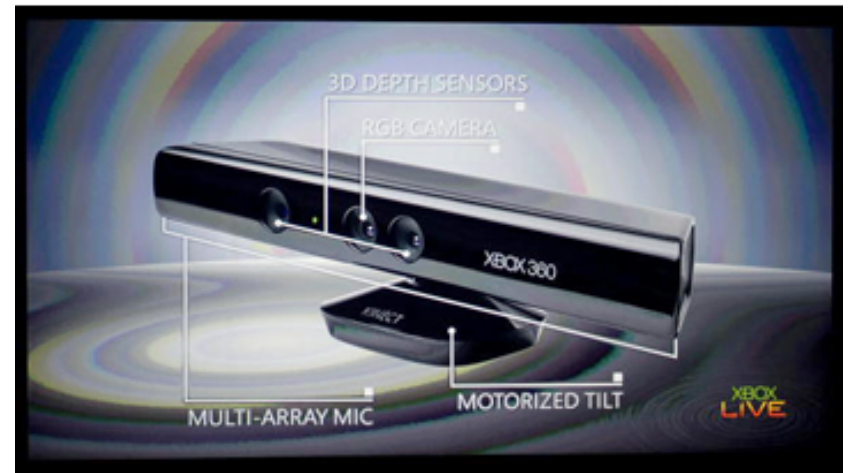
- Low Level Vision
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 - Deep understandings
 - Pose estimation



Face detection

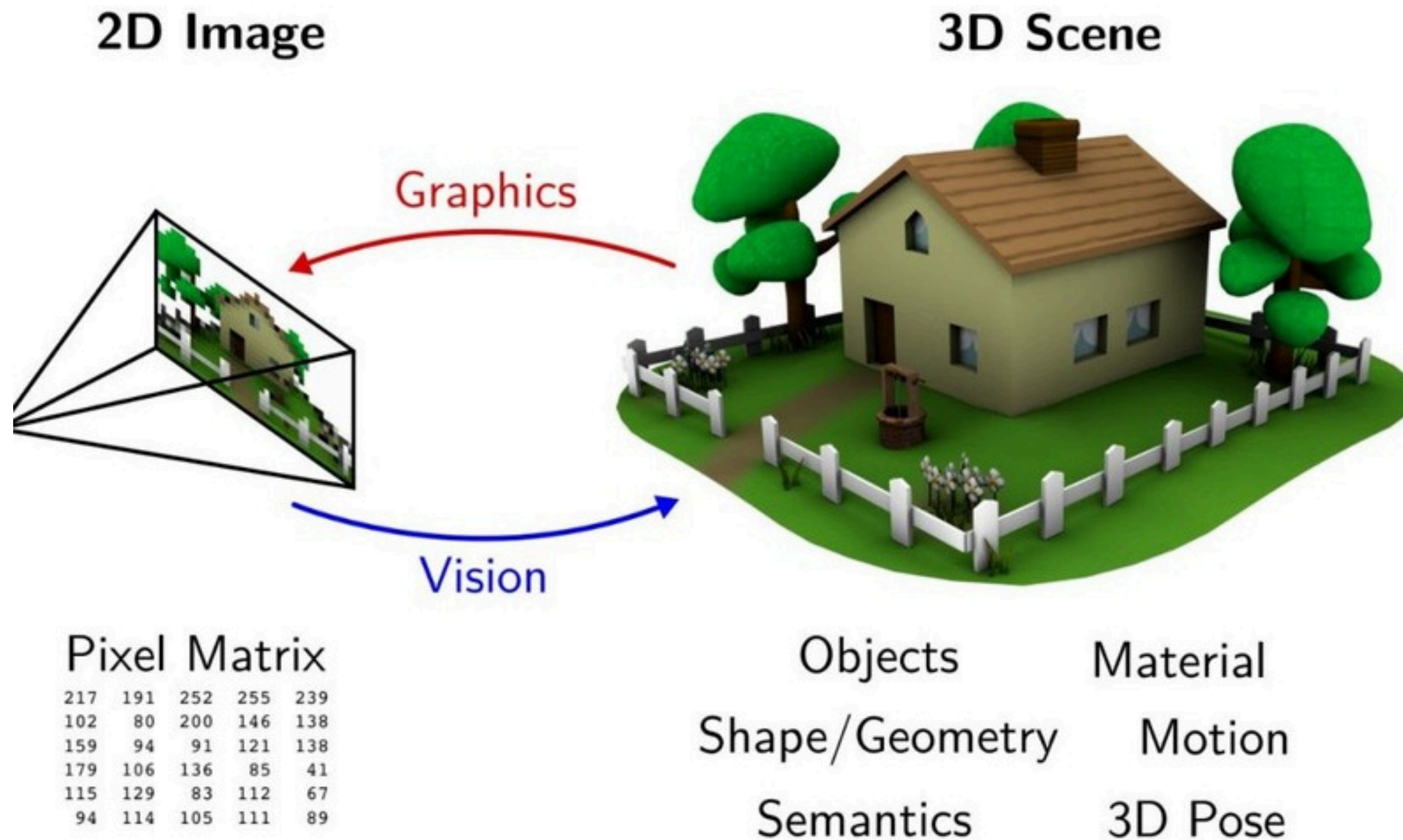


Vision-based interaction: Xbox Kinect



Why vision is so hard?

But why is computer vision so hard?



It is an ill posed problem

Challenges 1: view point variation



Challenges 2: illumination



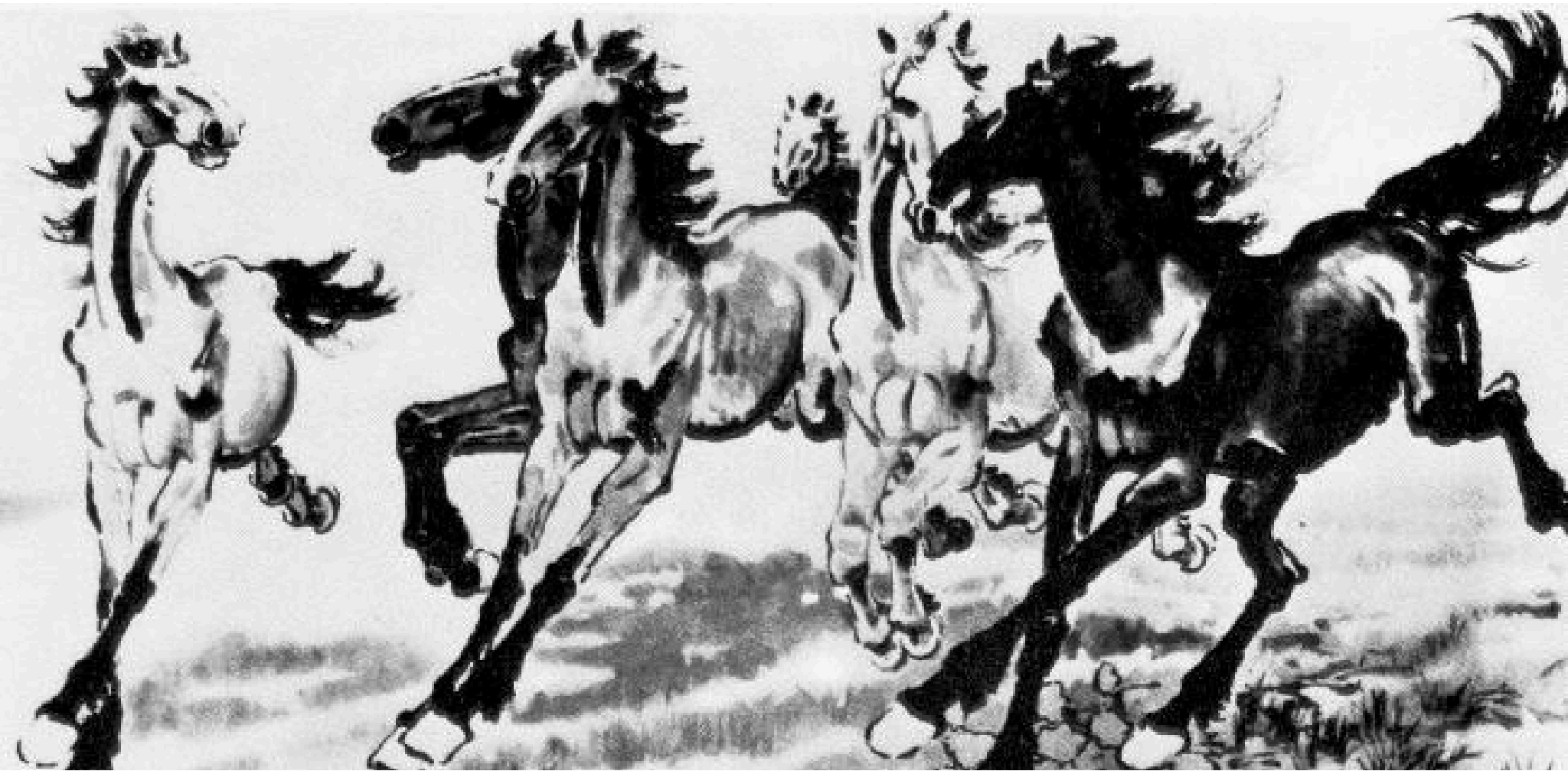
Challenges 3: occlusion



Challenges 4: scale



Challenges 5: deformation



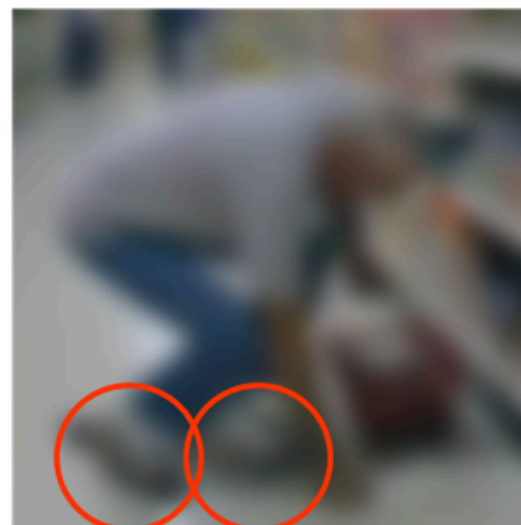
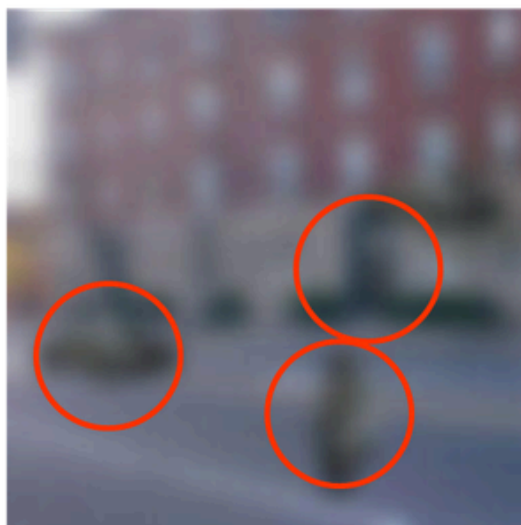
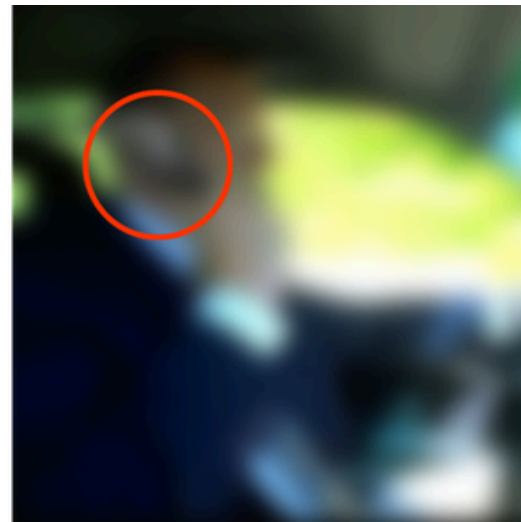
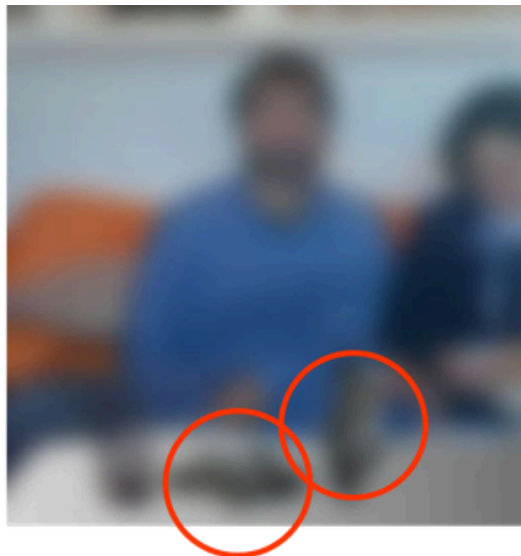
Challenges 6: background clutter



Challenges 7: object intra-class variation



Challenges 8: local ambiguity



Challenges 9: the world behind the image

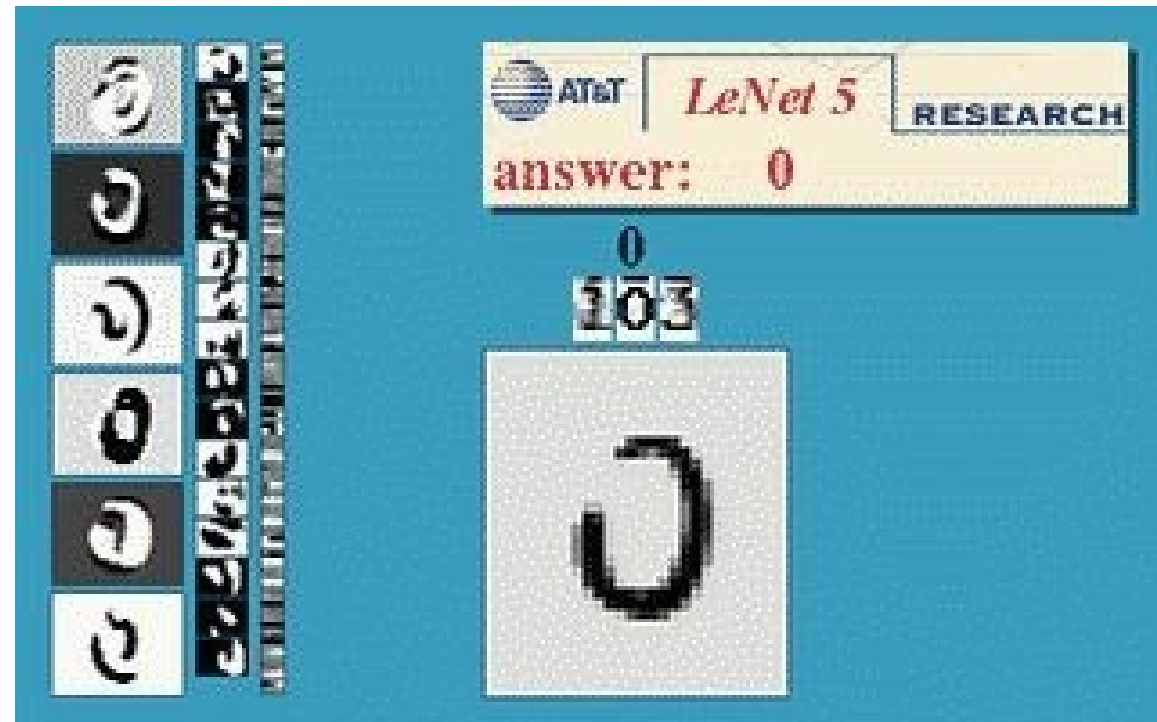


What Works Today?

Optical character recognition (OCR)

Technology to convert scanned docs to text

- If you have a scanner, it probably came with OCR software



Digit recognition, AT&T labs



License plate readers

http://en.wikipedia.org/wiki/Automatic_number_plate_recognition

Biometrics

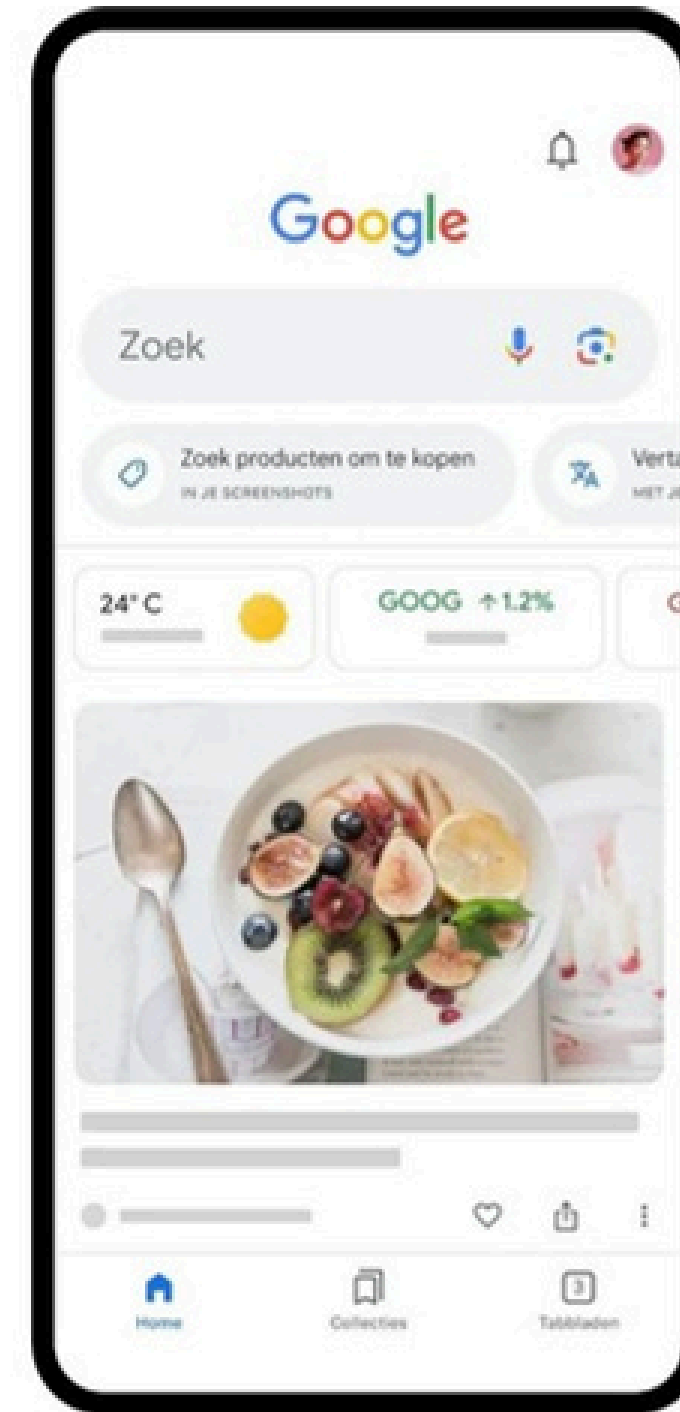
Fingerprint scanners on many new laptops, other devices



iPhone Face ID



Image Search



Face detection



Object recognition (in supermarkets)





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Last Updated: Wednesday, 31 August 2005, 05:44 GMT 06:44 UK

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Computer alert for drowning girl

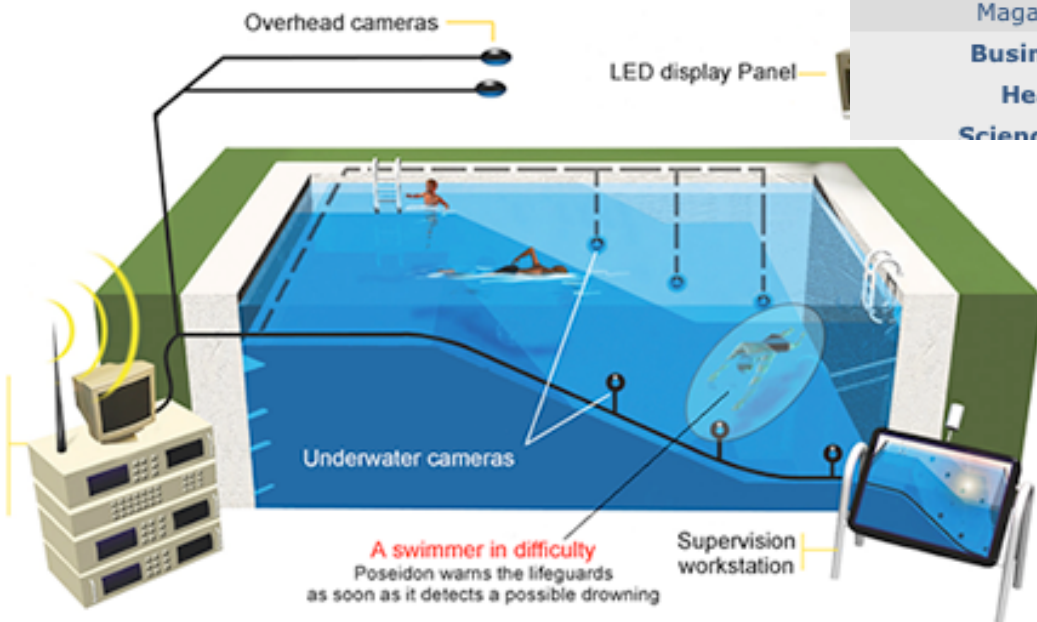
A 10-year-old girl has been saved from drowning by a computer system designed to raise the alarm when swimmers get into difficulties.



[VIDEO](#) **Watch the rescue**

The girl, from Rochdale, was at the deep end of the pool in Bangor, north Wales, when she sank to the bottom.

The £65,000 system, called Poseidon, detected her on the pool floor and sounded the alarm. A lifeguard pulled her out and she recovered in hospital.



Security

Local 

Cameras help confirm Scott suicide ruling

Friday, December 04, 2009



TAGS: [local](#), [paul meincke](#)

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Paul Meincke

More: [Bio](#), [News Team](#)

December 4, 2009 (CHICAGO) (WLS) -- Chicago police have closed the case in the death of Chicago School Board President Michael Scott.

Police Supt. Jody Weis says investigators used police cameras in the city to trace Scott's last steps in the hours before his body was found in November.

Scott's death has been ruled a suicide. The medical examiner's office concluded --not long after Scott's body was found -- that he had committed suicide. Police did not dispute the finding but wanted to pursue all the investigative leads they could. They say they have done that and have now reached the same conclusion.

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News Headlines 

Video

abcNEWS

- 2 suspects arrested in volleyball star's murder 47 min ago
- BP Gas Recall: BP finds, fixes source of bad gas
- Teachers union, board resume negotiating
- Back to School
- 5 injured in South Side shooting 49 min ago
- Pastor: Stacy Peterson said she lied for Drew



Vision-powered toys and robots in the 2020s



Scout home security robot - monitors your house



Unitree Go1's companion robot - like a dog



Automotive safety

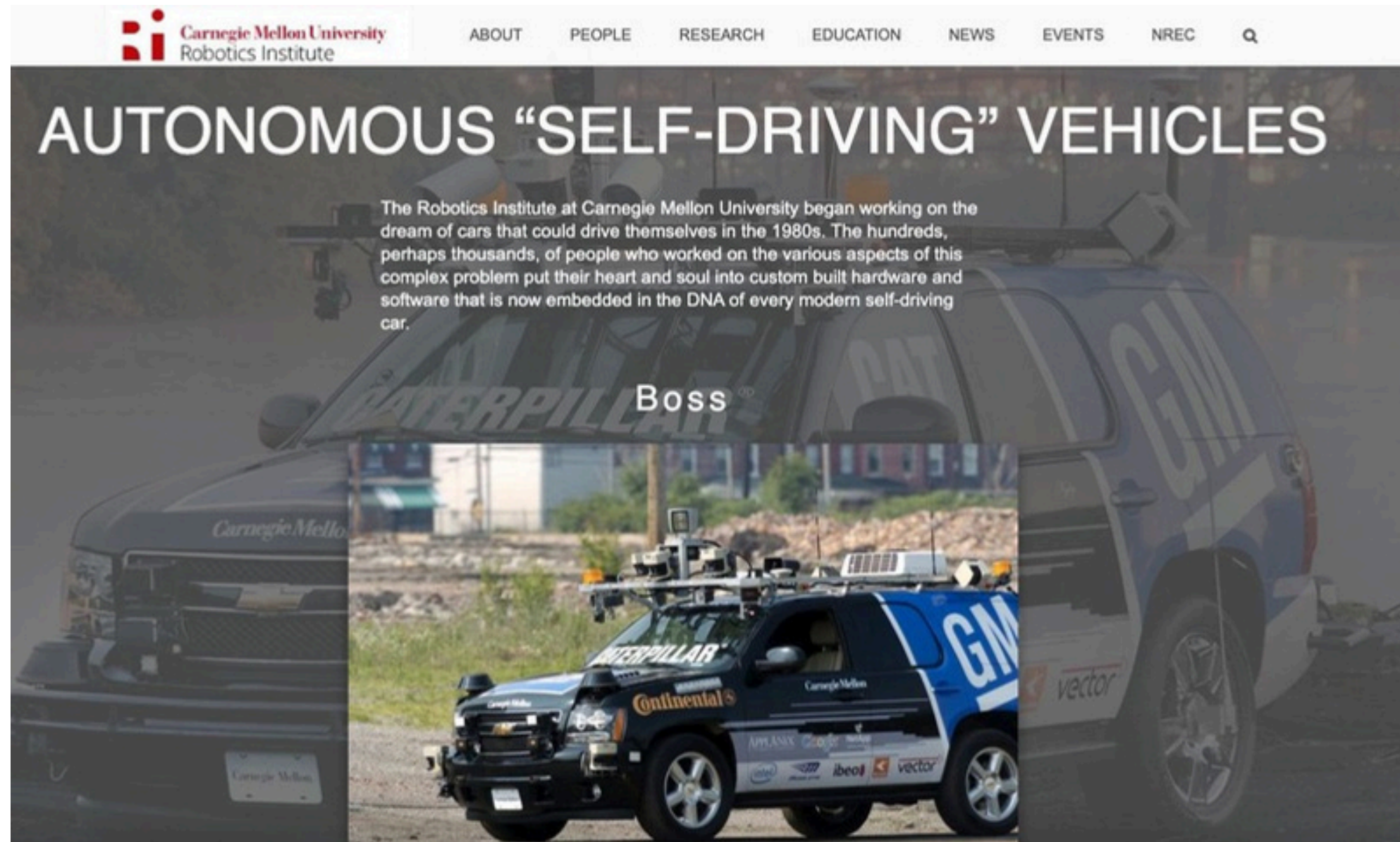
The screenshot displays the Mobileye website with a top navigation bar containing 'manufacturer products' and 'consumer products'. The main header reads 'Our Vision. Your Safety.' and features a top-down view of a car with yellow cones indicating the fields of view for 'rear looking camera', 'side looking camera', and 'forward looking camera'. Below this, three product sections are shown: 'EyeQ Vision on a Chip' with an image of a chip, 'Vision Applications' showing a pedestrian on a crosswalk, and 'AWS Advance Warning System' with a circular display showing a car icon and the number '0.8'. Each section has a 'read more' link. On the right side, there are two vertical panels: 'News' with headlines about Volvo's collision warning system and a link to 'all news', and 'Events' with links to 'Mobileye at Equip Auto, Paris, France' and 'Mobileye at SEMA, Las Vegas, NV', also featuring a 'read more' link.

- [Mobileye](#): Vision systems in high-end BMW, GM, Volvo models
 - Pedestrian collision warning
 - Forward collision warning
 - Lane departure warning
 - Headway monitoring and warning

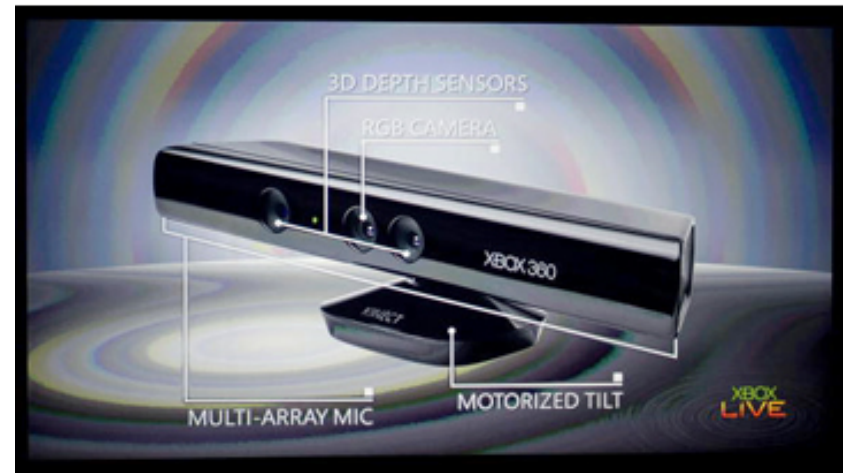
Google cars



At the end of the day, vision is for doing



Vision-based interaction: Xbox Kinect



Augmented reality, consumer products



Special effects: shape and motion capture



Vision for robotics, space exploration

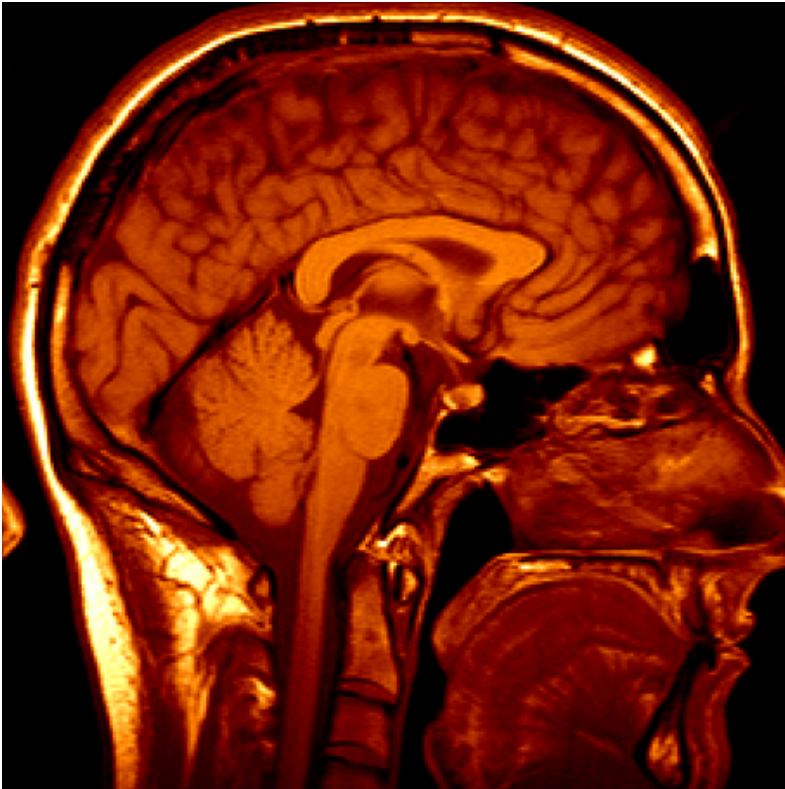


[NASA'S Mars Exploration Rover Spirit](#) captured this westward view from atop a low plateau where Spirit spent the closing months of 2007.

Vision systems (JPL) used for several tasks

- Panorama stitching
- 3D terrain modeling
- Obstacle detection, position tracking

Medical imaging



3D imaging
MRI, CT



Image guided surgery

Computer vision research in healthcare



assisted living, patient monitoring
[Lan et al, PAMI 2012]



autism screening

<http://www.gatech.edu/newsroom/release.html?nid=60509>

Augmented Reality

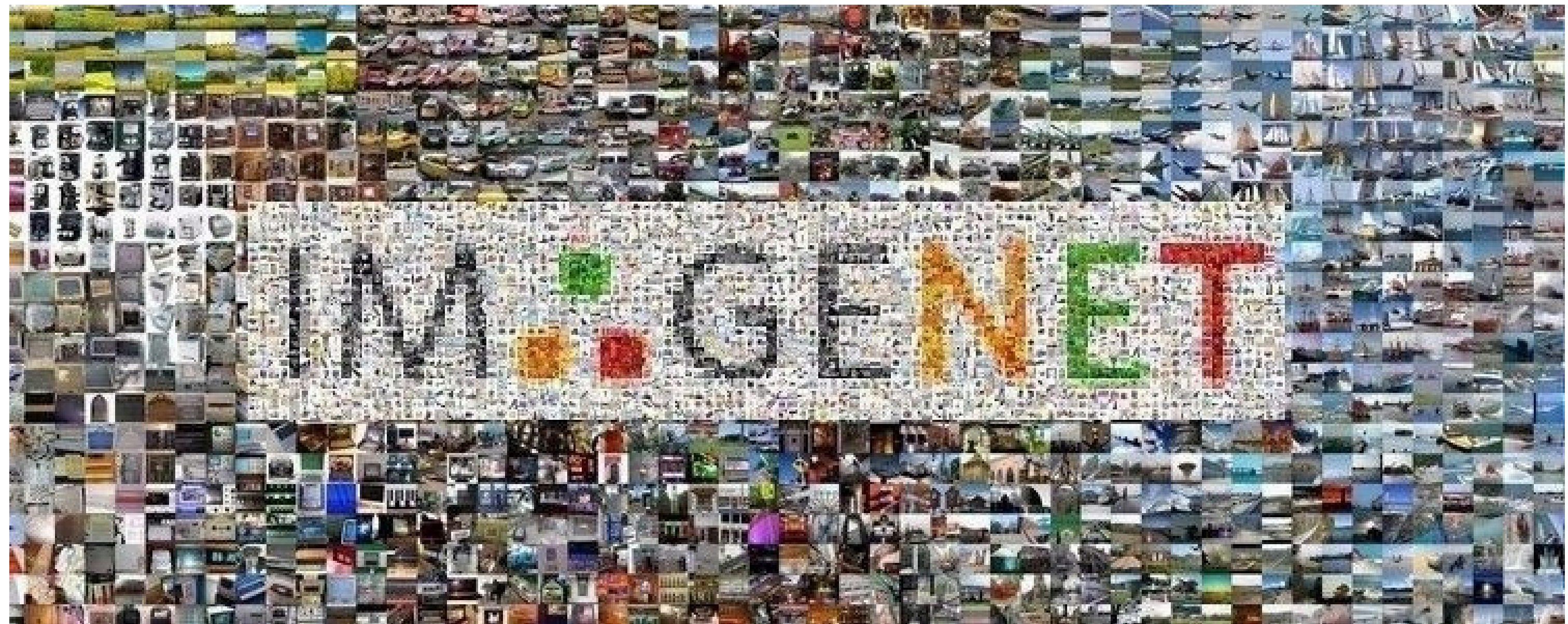


Apple Vision Pro, Snapstacles and Meta Quest



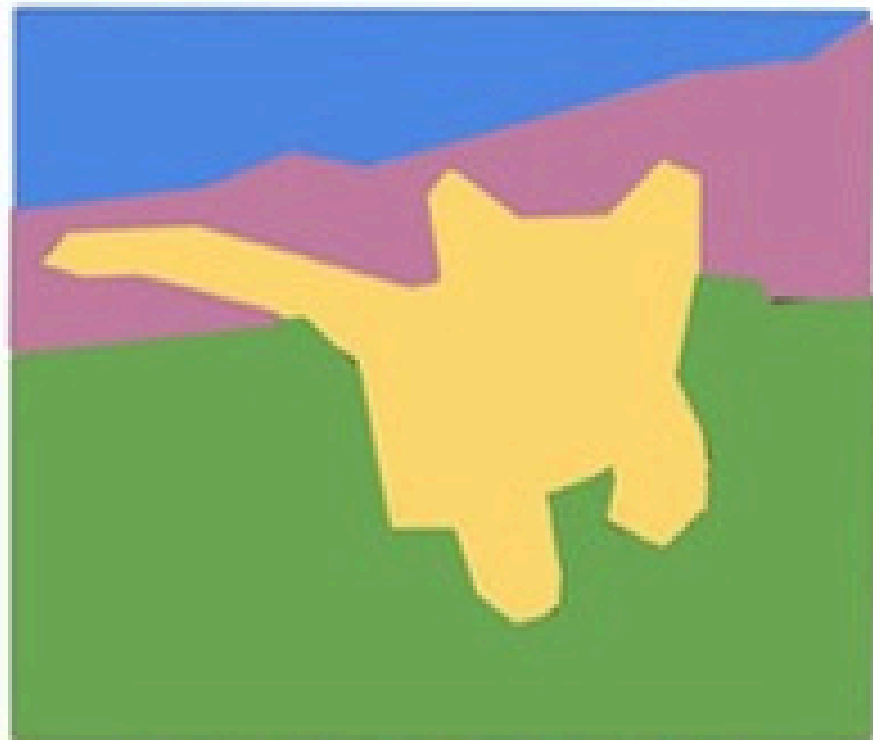
With smaller cameras and larger storage,

We began curating large scale databases of images online



With those images, we now train models to understand what is in an image

Semantic Segmentation



GRASS, CAT,
TREE, SKY

No objects, just pixels

**Classification
+ Localization**



CAT

Single Object

**Object
Detection**



DOG, DOG, CAT

Multiple Object

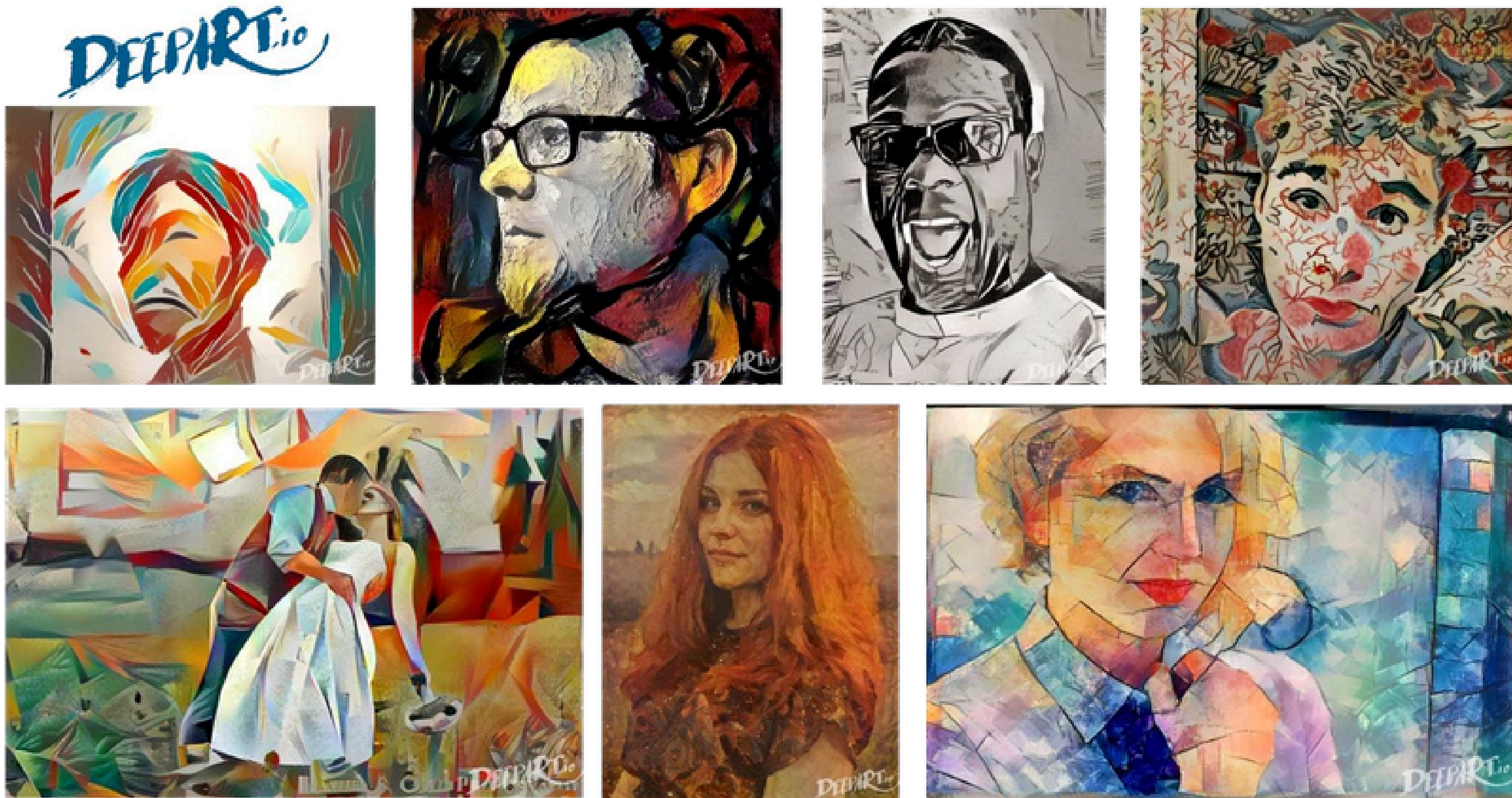
**Instance
Segmentation**

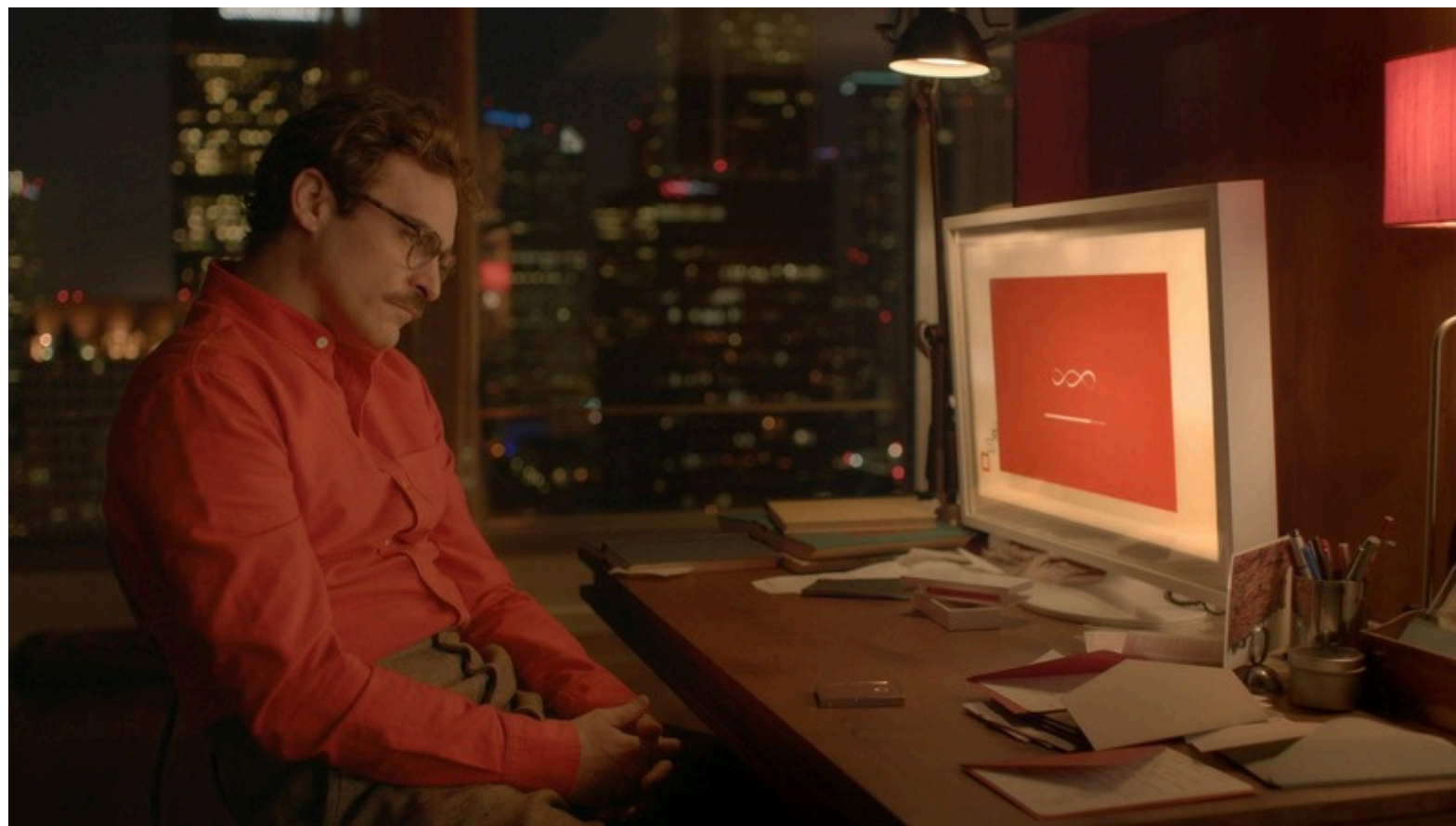


DOG, DOG, CAT

This image is CC0 public domain

We can also train models to **generate new images**





Why study computer vision?

Vision is useful: Images and video are everywhere!



Google™
Image Search



Google Photos

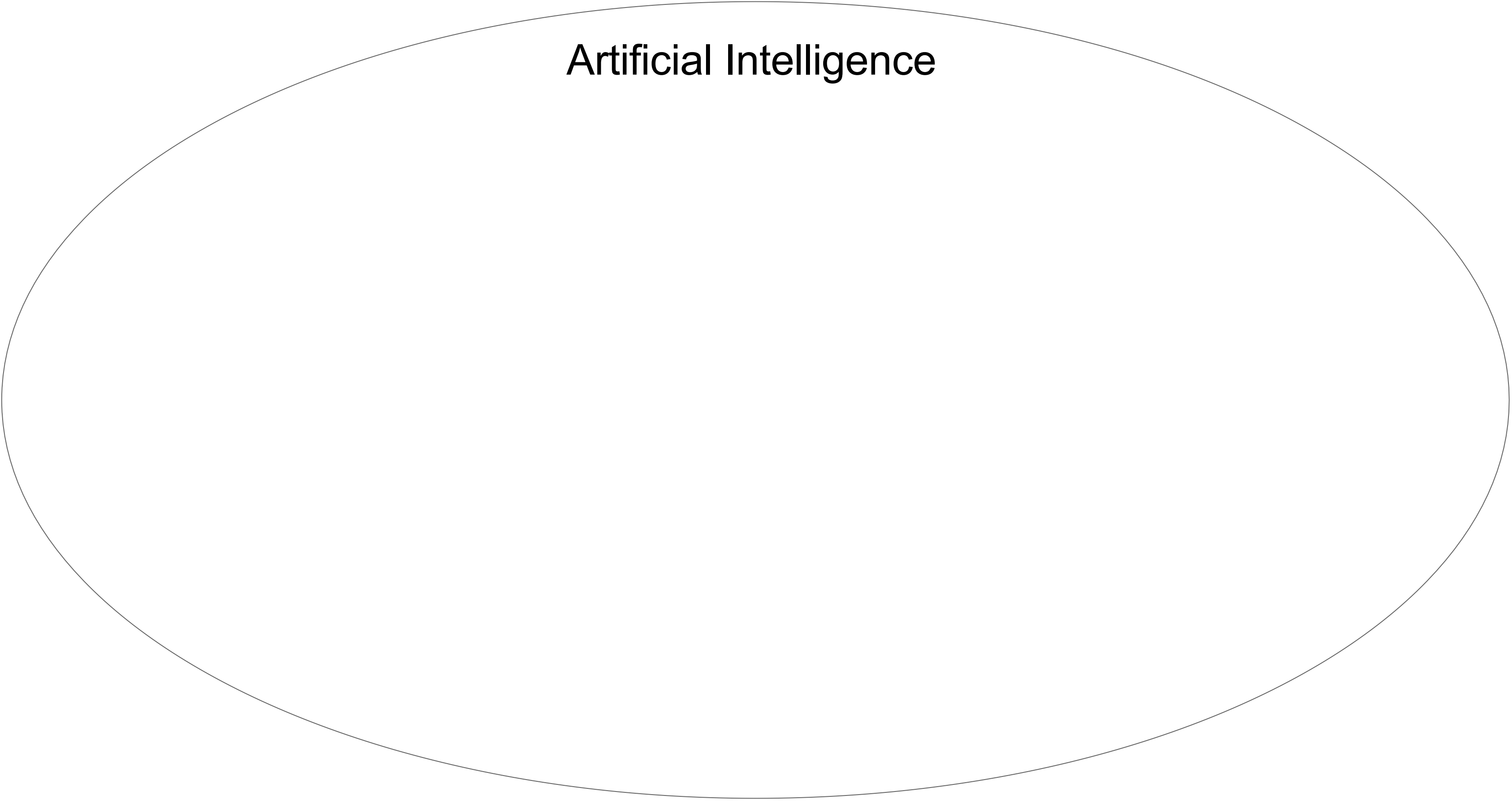
flickr™

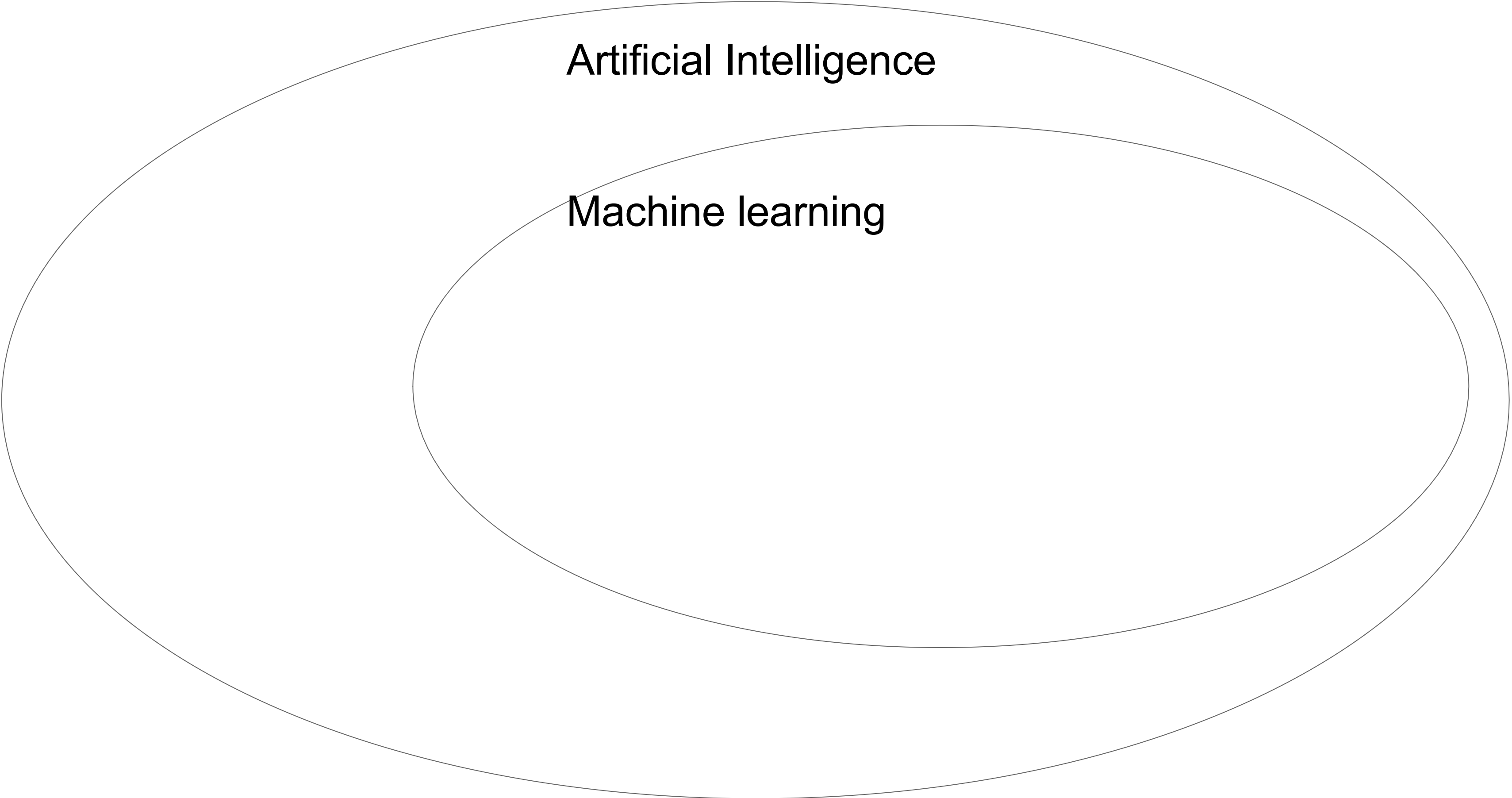
webshots™ beta

picsearch™

You Tube™
Broadcast Yourself™

Artificial Intelligence

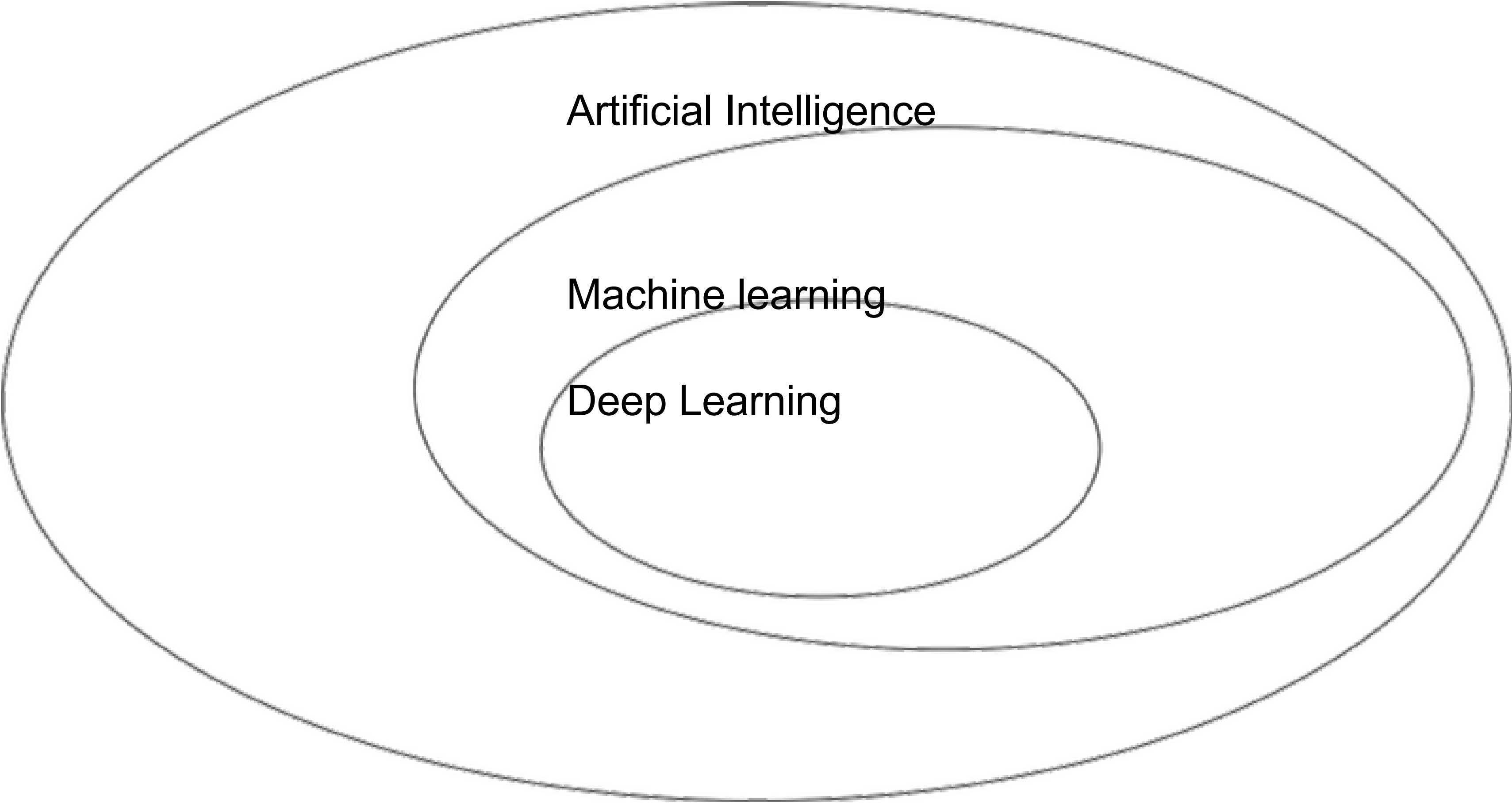
A large, empty oval shape with a thin black border, centered on the page. The text "Artificial Intelligence" is positioned at the top center of the oval.



A Venn diagram consisting of two concentric ellipses. The outer ellipse is larger and contains the text 'Artificial Intelligence'. The inner ellipse is smaller and is positioned within the outer one, containing the text 'Machine learning'. This visualizes that machine learning is a subset of artificial intelligence.

Artificial Intelligence

Machine learning

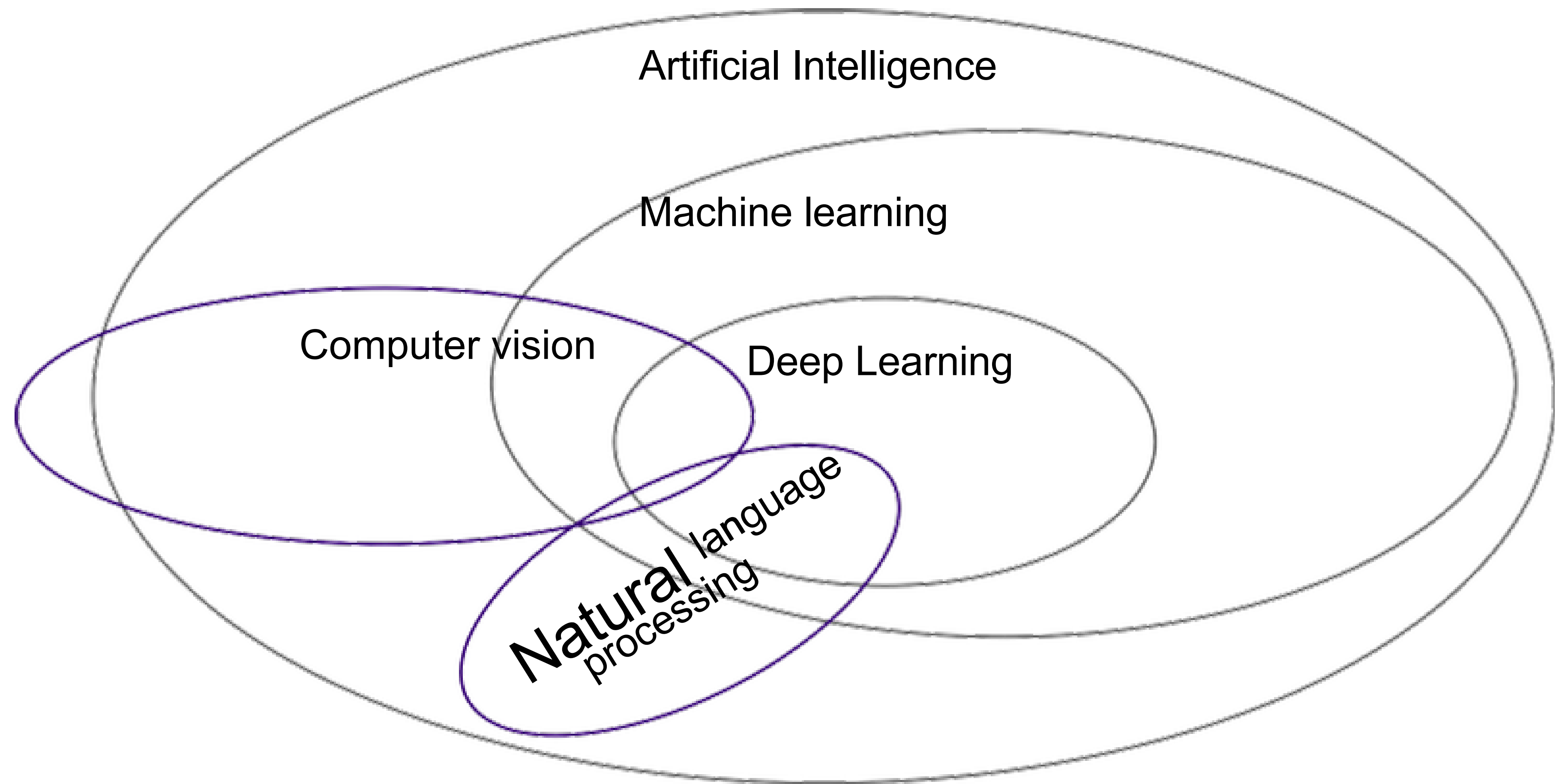


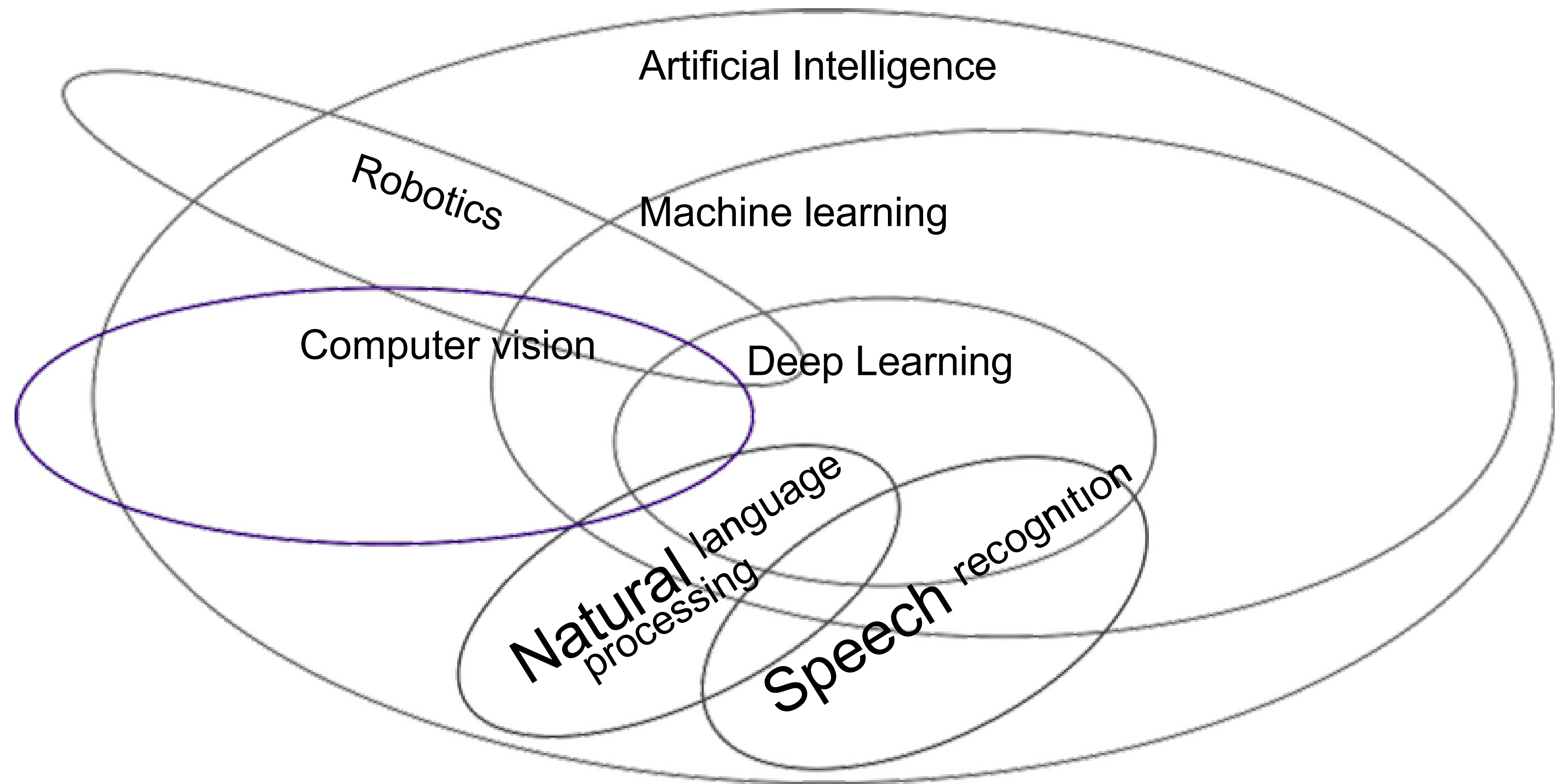
A Venn diagram consisting of three concentric ellipses. The outermost ellipse is labeled 'Artificial Intelligence'. Inside it is a medium-sized ellipse labeled 'Machine learning'. Inside the 'Machine learning' ellipse is the innermost, smallest ellipse labeled 'Deep Learning'. This illustrates that Deep Learning is a subset of Machine Learning, which is a subset of Artificial Intelligence.

Artificial Intelligence

Machine learning

Deep Learning





CSE 455 course
Artificial Intelligence

Robotics

Machine learning

Computer vision

Deep Learning

Natural language
processing

Speech recognition

Computer Vision and Pattern Recognition Conference (CVPR)



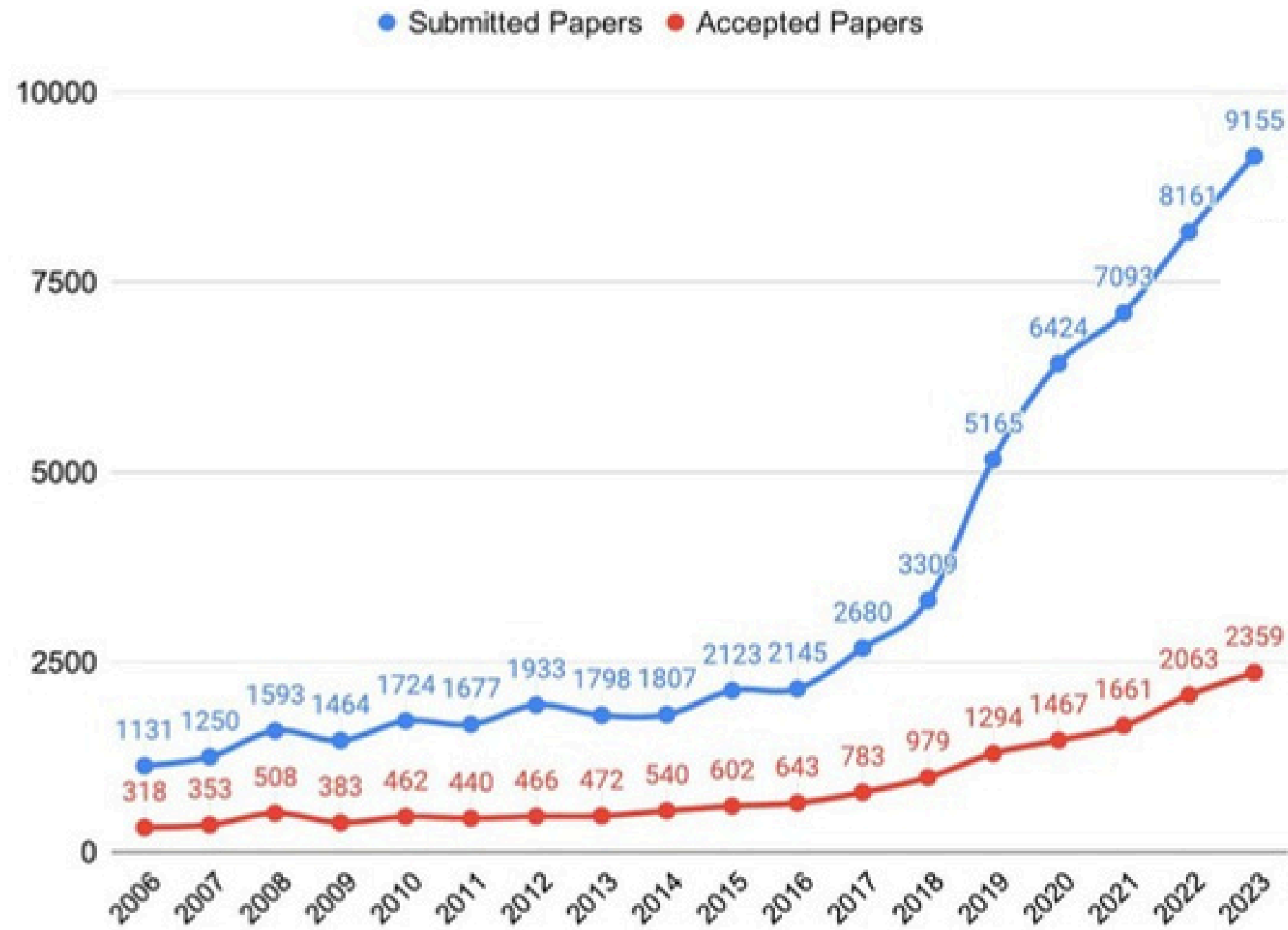
<https://cvpr.thecvf.com/>

Why should you go to CVPR? It is ranked #4 amongst all scientific publications **across all disciplines**

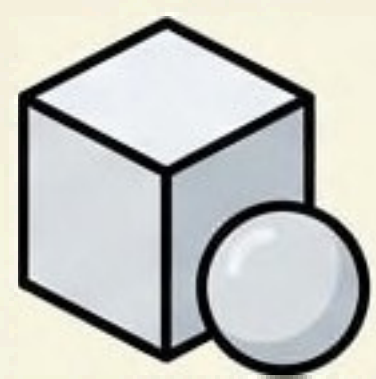
Publication	h5-index	h5-median
1. Nature	467	707
2. The New England Journal of Medicine	439	876
3. Science	424	665
4. IEEE/CVF Conference on Computer Vision and Pattern Recognition	422	681
5. The Lancet	368	688
6. Nature Communications	349	456
7. Advanced Materials	326	415
8. Cell	316	503

Source: [Google scholar](#)

CVPR has seen a large number of deep learning people enter



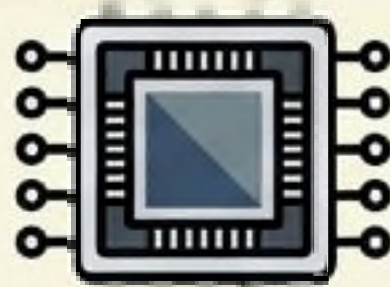
CVPR



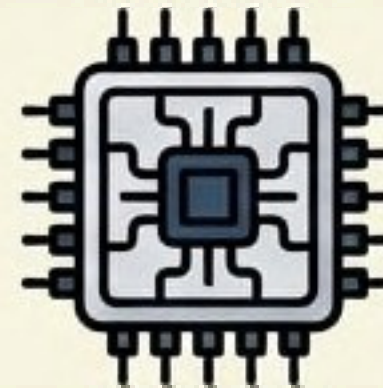
Scene



Lens



Sensor



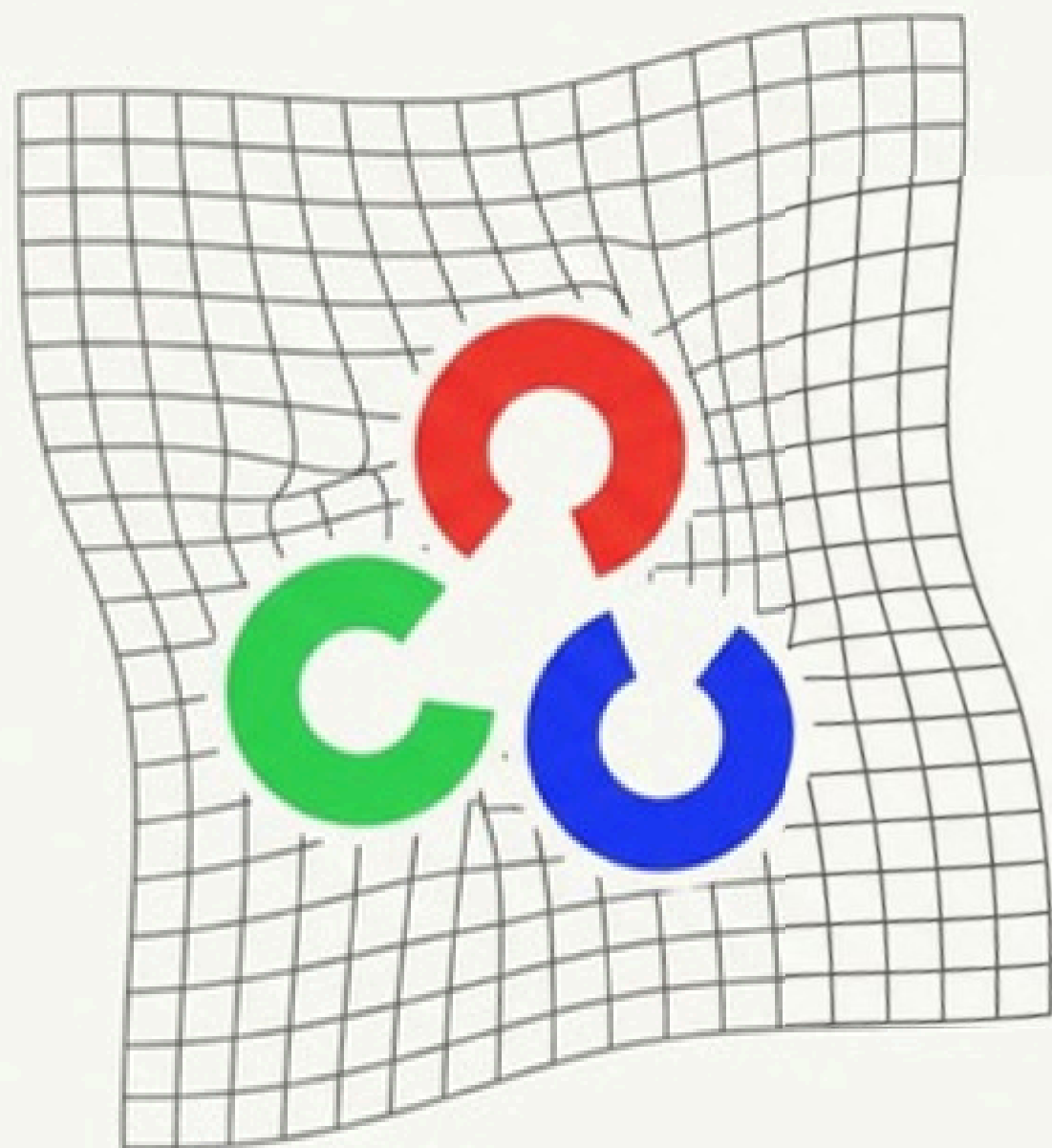
ISP



Computer Vision

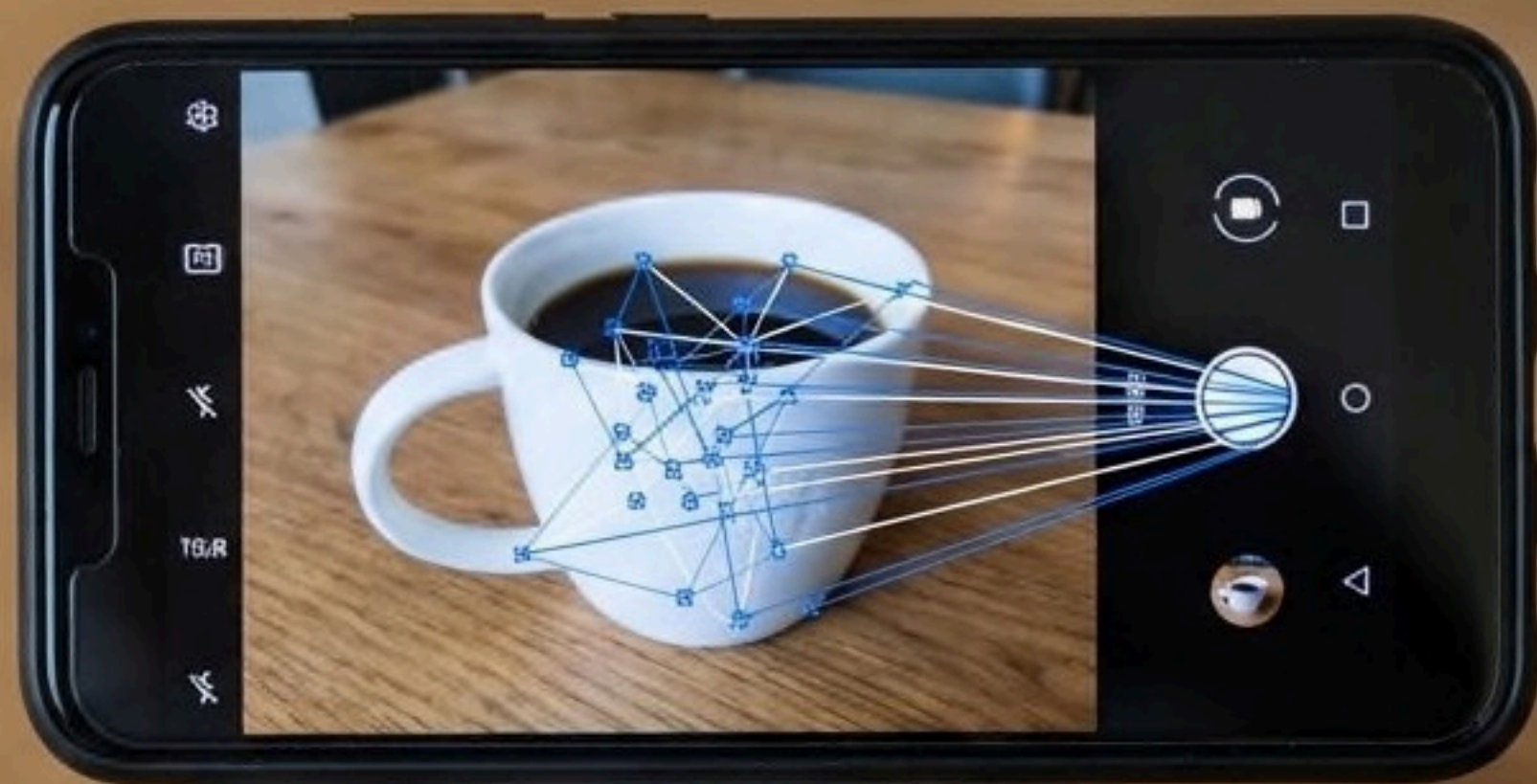
視覺系統的管線





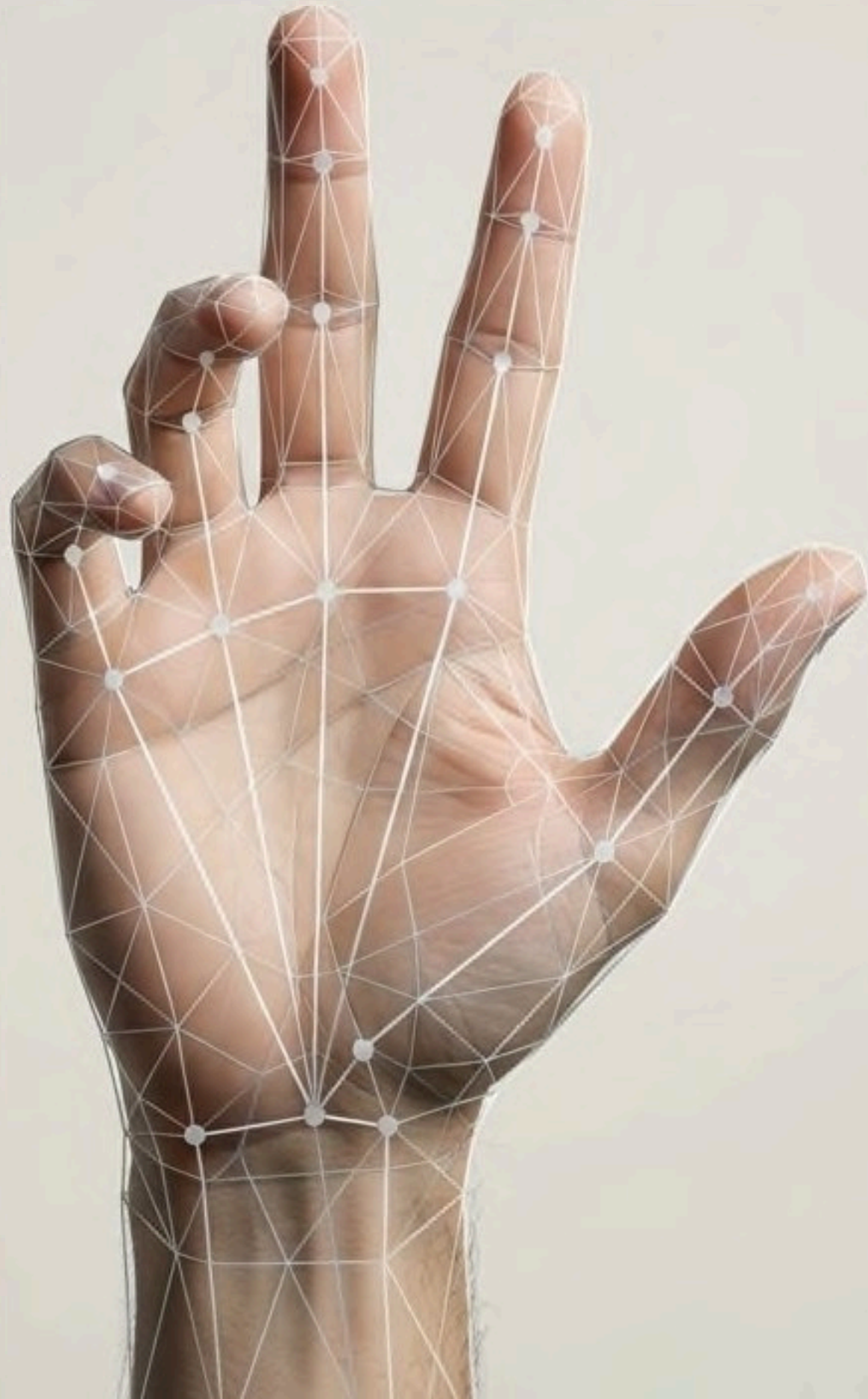
Part 1：影像處理基礎

- 課程介紹與 Computer Vision 系統
- Camera 與影像形成
- Pixel Processing
- Geometry Transformation
- Color Processing



Part 2 : Android Vision

- Android Vision Environment
- Feature Extraction
- Feature Matching 與 Tracking



Part 3 : ML Vision

- Machine Learning for Vision
- Deep Learning for Vision
- Face Detection / Landmark
- Hand Tracking / Gesture
- ML Kit Vision API
- Final Project

- 平時出勤與課堂作業 60%
- 期中考 20%
- 期末專題 20%

授課教師辦公室(EN306)

週三 13:00~14:00

Email georgeh@yuntech.edu.tw

建立Line群組

統計時間

有寫過Python嗎?

有用過Google Colab嗎?

有用過OpenCV嗎?

Colab + OpenCV環境說明

- 只要一個google帳號，就可以開始影像處理電腦視覺的學習

#Colab資料夾設定方法

```
#!pip install google.colab #如未安裝取消註解後執行
```

```
import os
```

```
import cv2
```

```
from google.colab.patches import cv2_imshow
```

```
from google.colab import drive
```

```
drive.mount('/content/drive')
```

##出現提示欄進行授權

```
os.chdir('/content/drive/My Drive/Colab Notebooks/Image Process') #切換該目錄
```

```
os.listdir() #確認目錄內容
```

```
# Check if the file exists before reading it.
```

```
# Print an error message and exit if it does not.
```

```
image_path = "Lenna.bmp"
```

```
if not os.path.exists(image_path):
```

```
    print(f"Error: Image file not found: {image_path}")
```

```
    exit()
```

```
image = cv2.imread(image_path, 0)
```

```
print(image.shape)
```

```
cv2_imshow(image)
```

儲存圖片

```
cv2.imwrite('resize.jpg', image)
```